

## RESEARCH ARTICLE

10.1029/2020JD033056

## Special Section:

Fire in the Earth System

## Key Points:

- Fire plume and climate modeling of a regional nuclear exchange finds potential global cooling, shorter in duration than previously assessed
- Aerosol emissions from fires and their representation in a climate model contribute uncertainty to the climate response
- A wide range of climate impacts are simulated depending on fuel availability at the detonation sites

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## Citation:

Wagman, B. M., Lundquist, K. A., Tang, Q., Glascoe, L. G., & Bader, D. C. (2020). Examining the climate effects of a regional nuclear weapons exchange using a multiscale atmospheric modeling approach. *Journal of Geophysical Research: Atmospheres*, 125, e2020JD033056. <https://doi.org/10.1029/2020JD033056>

Received 6 MAY 2020

Accepted 30 SEP 2020

Accepted article online 23 NOV 2020

## Examining the Climate Effects of a Regional Nuclear Weapons Exchange Using a Multiscale Atmospheric Modeling Approach

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**Abstract** Recent studies examine the potential for large urban fires ignited in a hypothetical nuclear exchange of one hundred 15 kt weapons between India and Pakistan to alter the climate (e.g., Mills et al., 2014, <https://doi.org/10.1002/2013EF000205>, and Reisner et al., 2018, <https://doi.org/10.1002/2017JD027331>). In this study, the global climate forcing and response is predicted by combining two atmospheric models, which together span the micro-scale to global scale processes involved. Individual fire plumes are modeled using the Weather Research and Forecasting (WRF) model, and the climate response is predicted by injecting the WRF-simulated black carbon (BC) emissions into the Energy Exascale Earth System Model (E3SM) atmosphere model Version 1 (EAMv1). Consistent with previous studies, the radiative forcing depends on smoke quantity and injection height, examined here as functions of fuel loading and atmospheric conditions. If the fuel burned is 1 g cm<sup>-2</sup>, BC is quickly removed from the troposphere, causing no global mean climate forcing. If the fuel burned is 16 g cm<sup>-2</sup> and 100 such fires occurred simultaneously with characteristics similar to historical large urban firestorms, BC reaches the stratosphere, reducing solar radiation and causing cooling at the Earth's surface. Uncertainties in smoke composition and aerosol representation cause large uncertainties in the magnitude of the radiative forcing and cooling. The approximately 4 yr duration of the radiative forcing is shorter than the 8 to 15 yr that has previously been simulated. Uncertainties point to the need for further development of potential nuclear exchange scenarios, quantification of fuel loading, and improved understanding of fire propagation and aerosol modeling.

**Plain Language Summary** If the detonation of nuclear weapons causes large fires, the smoke emissions could block sunlight and affect the global climate. A commonly studied scenario is the climate impact that would be caused by the detonation of one hundred 15 kt nuclear weapons in a “regional nuclear exchange” between India and Pakistan (Mills et al., 2014, <https://doi.org/10.1002/2013EF000205>, Reisner et al., 2018, <https://doi.org/10.1002/2017JD027331>). We simulate the global climate impacts of this scenario using new models for predicting the fire plume and climate and find that, when smoke from the fires remains in the lower troposphere, it is quickly removed and the climate impact is minimal. Conversely, when fires inject smoke into the upper troposphere or higher, more smoke is transported to the stratosphere where enough light is blocked to cause global surface cooling. Our simulations show that the smoke from 100 simultaneous firestorms would block sunlight for about 4 yr, instead of the 8 to 15 yr predicted in other models. Climate impacts are also shown to be sensitive to assumptions about the composition of the smoke. Additionally, we show that the global effects of the fires are sensitive to fuel availability and consumption, factors which are uncertain for cities in India and Pakistan.

## 1. Introduction

The potential effects of nuclear war on the Earth's climate have been examined using numerical models since the early 1980s. The fundamental question is whether smoke from terrestrial fires ignited by nuclear weapons would be injected or transported to the stratosphere, where it has the potential to alter radiative forcing and destroy stratospheric ozone (cooling the Earth's surface and increasing ultraviolet radiation) for a prolonged period of time.

The modeling tools brought to this problem have evolved over the decades, from a two-dimensional atmospheric chemistry model (Crutzen & Birks, 1982) and a one-dimensional radiative-convective equilibrium

model (Turco et al., 1983), to early Global Climate Models (GCMs Covey et al., 1984), and finally to modern Earth system models (e.g., Mills et al., 2014). Even as models have become more sophisticated and complete, no single modeling tool has been able to cover the full scope and complexity of the entire problem. Instead, models are applied selectively with the remaining pieces of the puzzle left to assumptions.

This piecewise modeling approach is necessary given the large number of complicated factors involved. An analysis of whether fires ignited by a nuclear war will cause global climatic and environmental consequences must address the following:

- The characteristics of the fires ignited by nuclear weapons (e.g., intensity, spread, and whether they generate sufficient buoyancy for lofting emissions to high altitudes); these are a function of many factors, including number and yield of weapons, target type, fuel availability, meteorology, and geography.
- The composition of the fire emissions (whether emissions include significant amounts of black carbon [BC] and organic carbon [OC] aerosols, and gases affecting atmospheric chemistry); these are a function of the fuel type, carbon loading, oxygen availability, and other factors.
- Whether the emissions are self-lofted by the absorption of solar radiation and to what heights; this is a function primarily of meteorology and particle size, composition, and absorption of solar radiation.
- The physical and chemical evolution of BC and other aerosol species in the stratosphere; this is a function of stratospheric chemistry and dynamics.

Initial studies (e.g., Turco et al., 1983) used available modeling tools and made assumptions as needed, to conclude that a massive exchange of tens of thousands of nuclear weapons by the United States and Soviet Union would affect the climate for decades, causing freezing temperatures, even in the summer, over much of the Earth (i.e., nuclear winter). Since the end of the Cold War, attention has shifted to the threat of regional nuclear wars which involve limited nuclear employment, especially in South Asia. There is no commonly accepted understanding of the numbers, yields, and types of weapons that might be employed and the associated targeting strategies. The climate modeling community has settled on a particular scenario, without broad agreement that it is the most plausible: an exchange of one hundred 15 kt weapons, all targeted on urban centers (Toon et al., 2007). The scenario also specifies the amount of BC injected into the upper troposphere (5 Tg), effectively prescribing the unknown factors in the first and second bullet points (above) and using numerical models to examine the third and fourth bullet points.

GCMs forced according to this regional nuclear exchange scenario simulate global surface cooling, peaking at  $\sim 1\text{--}1.5$  K about 3 to 4 yr after the war, with complete recovery taking 10 yr or more (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013). Additionally, models simulate stratospheric ozone reduction (Mills et al., 2008, 2014; Stenke et al., 2013) and increased ultraviolet radiation at the surface, which, in combination with the reduced growing season, have been hypothesized to cause global famine (Mills et al., 2014).

Reisner et al. (2018) also provide an assessment of the same one hundred 15 kt nuclear exchange, finding greatly reduced global climate impacts. The Reisner et al. (2018) approach deviates from previous efforts by modeling aspects of all four bullet points above: They use a coupled fire spread and atmospheric dynamics model to simulate fire ignition, fire dynamics and spread, and atmospheric dynamics including transport and dispersion of the smoke plume. The modeled fire uses fuel loading data from a combination of databases for suburban Atlanta, Georgia, an idealized atmosphere prescribed as dry and weakly stable, and an idealized wind profile. The Reisner et al. (2018) simulations result in conflagrations, rather than firestorms, and less BC (3.7 vs. 5.0 Tg per 100 fires). The BC plume resulting from the fire simulation is distributed throughout the troposphere, instead of in the upper troposphere only. The combination of injecting less BC, lower in the troposphere, results in less than 0.5 K surface cooling in an Earth system model.

Motivated by the different conclusions that have been reached for this scenario, we make our own assessment, which also uses numerical models to address aspects of all four factors bulleted above. The goal of this work is to model realistic spatial distributions of smoke from fires over Indian and Pakistani cities and to understand the climate response that would result from 100 simultaneous mass fires. The disparate scales, from individual urban fires (microscale) to the spread and climate effects of BC (global), require a multiscale modeling approach using both a weather and climate model. The emission and initial transport of smoke plumes is simulated using the Weather Research and Forecasting model (WRF Skamarock et al., 2019). WRF simulations use a parameterized fire at the surface with prescribed fluxes of heat, smoke, and moisture,

which are informed by historical firestorms and also set to match the quantities used in previous studies (i.e., 5 Tg BC is emitted in the base case). Ambient atmospheric conditions, including stability, wind, and moisture, play an important role in the dynamics of the fire and smoke plume, yet these local meteorological effects are not represented using a spatially uniform assumed smoke injection (Mills et al., 2014), or when simulations use an idealized dry atmosphere (Reisner et al., 2018). Here, we simulate emissions in meteorological conditions that are more characteristic of India and Pakistan. BC from the three-dimensional WRF emissions is then injected as a source term into the Energy Exascale Earth System Model (E3SM) atmosphere model Version 1 (EAMv1 Rasch et al., 2019). EAM simulates the self-lofting of BC by solar heating, its global transport, radiative forcing, stratospheric heating and ozone destruction, and finally, its removal. The BC source term used in previous studies to initialize GCMs (Mills et al., 2014; Reisner et al., 2018) is also used in EAM here, allowing for direct comparison of EAM to previously used models.

Our approach requires assumptions and making novel use of existing models, the consequences of which we evaluate using additional simulations. We perform a series of sensitivity studies examining plume rise within the WRF model, to establish that it behaves as expected to perturbations of atmospheric water vapor content and wind speed around climatological values. We also test the sensitivity of the fire simulations to changes in the assumed fuel loading for Indian and Pakistani urban centers. The fuel loading is assumed to be  $16 \text{ g cm}^{-2}$  in our base case simulations (section 2.1.1), but Reisner et al. (2018) uses a spatially varying value averaging less than  $1 \text{ g cm}^{-2}$  (Reisner et al., 2019). Therefore, we perform a set of simulations with lower fuel loadings and thus lower emitted quantities of BC. Additionally, we test the sensitivity of the climate forcing and response to uncertainties in the physical and chemical properties of BC, by injecting it into different modes of the aerosol module in EAM and by including particulate organic matter (POM) as well as BC.

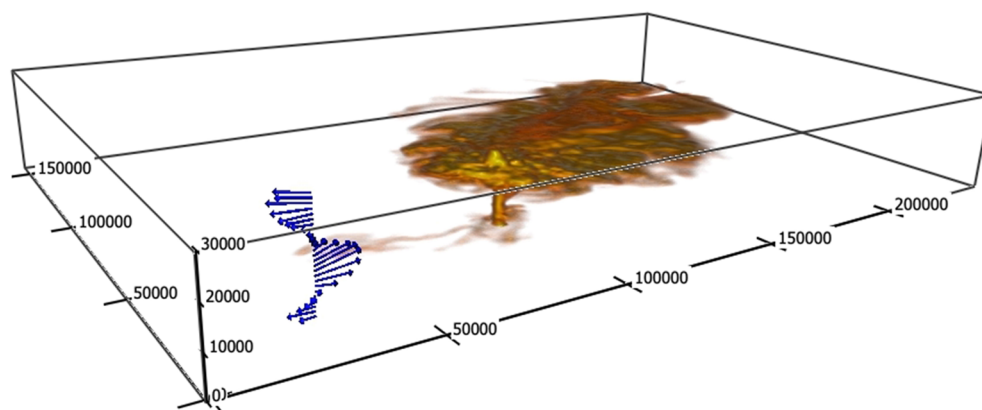
The base case fires simulated in WRF using site-specific local meteorology from India and Pakistan result in relatively high injection heights reaching into the stratosphere, thus higher than previously considered tropospheric injections (Mills et al., 2008, 2014; Pausata et al., 2016; Reisner et al., 2018; Robock et al., 2007; Stenke et al., 2013). Injected as a source term into EAM, the smoke from these fires reduce both solar radiation at the surface and stratospheric ozone. The radiative forcing is significant and causes global surface cooling, but ranges widely according to the characteristics of injected BC and whether POM is additionally included in the smoke plume. The duration of forcing and response are universally shorter in EAM than in the GCMs used in the aforementioned studies. Furthermore, we establish that fuel loading is a critical and uncertain parameter. Use of lower fuel loadings results in no significant global impacts, which may explain the difference between the findings of Reisner et al. (2018) and previous studies (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013).

## 2. Methodology

### 2.1. Idealized Fire Simulations

The Weather Research and Forecasting model is a regional numerical weather prediction model used for both operational weather forecasting and research studies (Skamarock et al., 2019). Advanced Research WRF (WRF-ARW) uses a non-hydrostatic dynamical core and can be applied to grid resolutions ranging from meters to tens of kilometers. In this work, we use WRF v4.0.1 in “ideal” mode to simulate individual fires, as depicted in Figure 1. Separate simulations are performed for fires in 20 urban areas (the 10 most populous in India and the 10 most populous in Pakistan) during the months of January and August, for a total of 40 fire simulations. A single atmospheric sounding is used to initialize each WRF simulation, and this sounding is extracted from an EAMv1 simulation (see section 2.2) at the detonation time and location of each fire, yielding a consistent atmospheric state between the fire and climate simulations. January detonations are consistent with Mills et al. (2014), Pausata et al. (2016), and Reisner et al. (2018). Additional fire simulations are also completed using a January climatological sounding with the goal of better understanding model sensitivity to background meteorology and fuel loading.

Although WRF has been coupled to a fire behavior model (WRF-FIRE Coen et al., 2013), the coupled capability has been developed for wildland fires, rather than urban fires, and is therefore not used here. Instead, the fire source is prescribed in the simulations by specifying surface fluxes. Smoke contains gases and aerosols which can modify atmospheric processes and dynamics. Regardless, smoke is frequently neglected or represented as a passive tracer when individual fires are simulated and the focus of the study is on fire spread or the atmospheric transport of smoke over a short timescale (Cunningham & Reeder, 2009; Coen et al., 2013;



**Figure 1.** Simulation of a prescribed fire in the WRF model with  $16 \text{ g cm}^{-2}$  fuel loading and January climatology from south central India. Colored volume rendering shows concentration of smoke emissions. Vectors indicate horizontal wind speed and direction. Axes are in meters. The visualization software used is VAPOR (Clyne et al., 2007).

Jimenez et al., 2018; Muoz-Esparza et al., 2018; Reisner et al., 2018). Alternatively, smoke emissions, and their interactions with the atmosphere, can be modeled in WRF using the coupled chemistry modules of WRF-Chem (Grell et al., 2005). The High Resolution Rapid Refresh (HRRR-Smoke) modeling system is an experimental forecast product based on the WRF model that represents smoke as a single prognostic variable and includes interactions of the smoke with atmospheric processes (Ahmadov et al., 2017). Kochanski et al. (2019) coupled the WRF-Fire and WRF-Chem capabilities, representing fire emissions as  $\text{SO}_2$ ,  $\text{PM}_{2.5}$ , organic carbon, black carbon, and  $\text{PM}_{10}$ . The study was able to model fire-smoke-atmosphere feedbacks, but also noted the preliminary nature of the work and the need for more research and validation. In this work, we choose to model smoke as a passive tracer in the WRF simulations of fire (i.e., WRF-Chem is not used).

### 2.1.1. Fire Source Term

The firestorms of World War II, such as those in the cities of Hamburg, Germany (27 July 1943), and Hiroshima, Japan (6 August 1945), are described as a mass fire in which there is a large central convective core into which street level winds flow from all directions (Carrier et al., 1985; Glasstone, 1962; Rodden et al., 1965). The radially converging winds near the surface prevent outward propagation of the fire, while nearly all combustible material within the area of ignition is consumed, distinguishing this type of fire from those where ambient winds aid in fire spread. A typical firestorm burns an area of several kilometers uniformly, with fire intensity typically peaking 2 to 3 h after bombing and subsiding after approximately 6 h (Carrier et al., 1985; Glasstone, 1962; Rodden et al., 1965). While the Hamburg firestorm was ignited by massive conventional and incendiary explosive bombing raid, the fire in Hiroshima was ignited by a nuclear weapon of approximately 15 kt yield (Holmes & White, 2013). Despite the difference in ignition source between the fires in Hiroshima and Hamburg, the incinerated area was about the same in the two fires (Rodden et al., 1965). The atomic bombing of Nagasaki, with a weapon of approximately 20 kt yield (Holmes & White, 2013), resulted in a fire with approximately one fourth the area of the fire in Hiroshima (Glasstone, 1962), providing a counterexample where use of a nuclear weapon did not create a firestorm. The reasons for this difference are given as limited fuel availability, the presence of fire breaks, and shielding of thermal radiation by topography (Glasstone, 1962).

The goal of the fire simulations in this work is to better characterize the spatial and temporal distribution of smoke from a mass urban fire resulting from a 15 kt nuclear detonation. Therefore, our modeling is informed by the Hiroshima firestorm, and the Hamburg firestorm, due to its rough similarity to the Hiroshima firestorm in size and duration. The assumption that all 100 detonations cause fires, and that these fires are more like the Hiroshima firestorm than Nagasaki, is a worst-case scenario. The studies of Penner et al. (1986) and Toon et al. (2007) also use fire parameters based on these historical cases (Hiroshima and Hamburg), so our fire parameters have the additional benefit of being similar to these previous studies. To produce simulations of fires similar to Hiroshima and Hamburg, it is assumed that the terrain is flat (i.e., topography does not provide shielding of thermal radiation) and there is uniform fuel loading over the area where thermal radiation is sufficient to ignite standard construction materials, such as wood. The WRF



**Table 1**  
*Prescribed Parameters for Fires in WRF Using Various Fuel Loadings for the Same Fire Area*

| Fuel loading ( $\text{g cm}^{-2}$ ) | Heat flux ( $\text{W m}^{-2}$ ) | Water vapor (Tg) | Black carbon (Tg) |
|-------------------------------------|---------------------------------|------------------|-------------------|
| 1                                   | $9.80 \cdot 10^3$               | 6.9              | 0.31              |
| 5                                   | $4.90 \cdot 10^4$               | 34.6             | 1.56              |
| 10                                  | $9.80 \cdot 10^4$               | 69.1             | 3.13              |
| 16 <sup>a</sup>                     | $1.57 \cdot 10^5$               | 111.5            | 5.0               |

*Note.* BC emissions are after assumed rainout. Emissions of water vapor and BC are summed over 100 fires; divide by 100 for emissions from an individual fire.

<sup>a</sup>Indicates the default value of fuel loading used, unless otherwise specified.

model source code is modified to allow for specification of surface fluxes of heat, water vapor and smoke (or black carbon), requiring quantification of these three fluxes, as well as the fire shape, size and duration.

The Hiroshima firestorm burned an area of about 11 to 13  $\text{km}^2$  in 4 to 9 h, taking 20 to 30 min to develop into a firestorm (Glasstone, 1962; Rodden et al., 1965). The Hamburg firestorm burned a comparable 12  $\text{km}^2$  in about 6 h (Carrier et al., 1985). Therefore, we specify a circular area with a 2 km radius (12.57  $\text{km}^2$ ) for our fires. Each fire has a 30 min ramp-up period as surface fluxes increase linearly from zero, followed by a 4 h fire duration where surface fluxes are constant. The 4 h duration is chosen because it is the shortest time estimate for the fire in Hiroshima, and releasing a given mass of emissions and burning a given fuel amount over the shorter time period will result in higher smoke concentrations and heat fluxes, thus providing a worst-case estimate.

Surface fluxes of heat are calculated using the estimated fuel loading, energy content of the fuel, and area burned. Fuel loading and energy content are the principal inputs into the calculation of the energy released by a fire, yet values of these parameters are highly uncertain. Schubert, a German fire chief, was commissioned to develop fuel loading numbers for two areas of fire in Hamburg, which were reported as 11.2 and 15.7  $\text{g cm}^{-2}$  (Schubert, 1969). Fuel loading in Hiroshima was not studied in detail, however, Rodden et al. (1965) estimates the fuel loading to be 3.9  $\text{g cm}^{-2}$  and uses this value to set a minimum fuel loading criteria for formation of a firestorm. A fuel loading value of 16  $\text{g cm}^{-2}$  is often used for the Hamburg fire (Penner et al., 1986; Toon et al., 2007), and we choose to use this as the default value in this study. Using  $15 \cdot 10^3 \text{ J g}^{-1}$  as the energy content of air-dried wood (Toon et al., 2007), this results in a surface heat flux of  $1.57 \cdot 10^5 \text{ W m}^{-2}$ . Moisture released in the fire is specified as 0.55 g water per 1 g of fuel combusted, as suggested in Cunningham and Reeder (2009), and based on the chemical reaction for the combustion of cellulose. This value is likely high for an urban fire, however, a sensitivity study (not shown) revealed little to no dependence of plume height on this value. Finally, we choose to release 5 Tg ( $5 \cdot 10^{12} \text{ g}$ ) BC into the climate model per 100 fires, for consistency with the studies of Mills et al. (2008, 2014), Robock et al. (2007), Stenke et al. (2013), Toon et al. (2007), and Pausata et al. (2016). Those studies use an emission of 6.25 Tg BC and assume 20% is removed by rainout during the plume rise, resulting in 5 Tg BC remaining in the atmosphere. BC removal is not modeled in WRF, so 5 Tg BC is emitted from the fire in WRF and remains in the atmosphere throughout the simulation. The BC emission factor required to emit 5 Tg BC is 2.49 g (BC)/g (fuel).

Toon et al. (2007) provides a detailed discussion of fuel loading in urban areas and develops a relationship between fuel loading and population density resulting in higher fuel loading estimates than are tested here, whereas Reisner et al. (2018) uses a lower fuel loading of approximately 1  $\text{g cm}^{-2}$  (Reisner et al., 2019). One possibility is that only a fraction of available fuel would burn. We do not perform our own assessment of fuel loading for urban centers in India and Pakistan, but we perform a subset of our analysis for the lower fuel loading values of 1, 5, and 10  $\text{g cm}^{-2}$  (see Table 1 for a summary of fire and emissions parameters).

### 2.1.2. Fire Simulation Setup

Idealized simulations of individual fires are performed using WRF with a two-domain nested setup where two-way coupling or feedback is turned on. Fire parameters are prescribed on the inner domain only as a circular area of surface fluxes of heat, moisture and smoke. Using two-way coupling between the nests, the solution on the outer domain is nudged toward the solution from the inner domain, thereby transferring the convective fire column and emissions to the outer domain. A horizontal grid resolution of 200 m is used on the inner domain and a resolution of 1,000 m is used on the outer domain. This two-domain setup is

chosen because it allows for adequate resolution of the fire on the inner domain, while the outer domain is large enough to completely contain the fire emissions, even after the 4.5 h duration of the fire. The size of the outer domain is based on the wind speed and direction of the initialization sounding and is hundreds of kilometers in the  $x$  and  $y$  dimensions. The inner domain is always chosen to have horizontal extents of 60 km by 60 km, while the extents of the outer domain and the nested position of the inner domain vary between simulations. The model top is 30 km, and 151 vertical levels are used. The grid is stretched such that the vertical resolution is approximately 98 m near the surface and 142 m near the tropopause. Rayleigh damping is used in the top 5 km of the domain. The time step is 0.25 s (outer domain) and 0.05 s (inner domain) to accommodate strong updrafts over the fire.

Atmospheric physics parameterizations include the Purdue-Lin microphysics (Chen & Sun, 2002) and RRTMG shortwave and longwave radiation schemes (Iacono et al., 2008). Sub-grid scale turbulence is represented by an eddy viscosity closure that solves a prognostic equation for turbulent kinetic energy (i.e., 1.5-order TKE-based closure). Surface stress is represented through inclusion of a drag term with  $C_d = 0.01$ . A land-surface model is not used, therefore surface fluxes of heat and moisture outside of the fire area are zero and considered negligible in comparison to the fire source term. The simulations occur on 1 January and 1 August of an arbitrary year at the latitude and longitude of each of the 20 detonation sites. Each WRF simulation begins at 4:30 UTC with a 30 min spin-up period, followed by a 30 min ramp-up time where the fire parameters increase linearly to the steady state values. The steady fire is simulated between 5:30 and 9:30 UTC. The time and location parameters are used in the Coriolis term, as well as in the radiation model.

WRF is initialized with an atmospheric sounding that includes horizontal velocities, potential temperature, and water vapor mixing ratio. These values are extracted from instantaneous EAM fields for the column containing the detonation site at 4:30 UTC. The WRF terrain is flat and is set to the elevation of the EAM grid cell at the detonation site. A subset of WRF simulations is initialized from an ERA-Interim (Dee et al., 2011) climatological sounding using a January climatology from south India, rather than an instantaneous sounding from EAM. In these cases, coordinates are set to 12.75°N, 76.5°E and the terrain height is set to 0 m or mean sea level. WRF simulations initialized with climatology are used in the sensitivity studies, while simulations for individual cities are initialized with instantaneous soundings from EAM, accounting for local meteorology.

The WRF idealized setup is similar those used in other studies of fire (Badlan et al., 2017, 2019; Cunningham & Reeder, 2009). Cunningham and Reeder (2009) use WRF at the same horizontal resolution (200 m), with similar physics options (microphysics and turbulence closure), and a parameterized fire source to simulate a 2003 Australian wildfire and associated pyroconvection. They note a qualitative comparison to the observed fire and that the modeled plume reached the observed height of the fire (14 km) when surface fluxes of both heat and water vapor were specified at the fire source. As an additional test of our setup we modeled the 1943 firestorm in Hamburg using a summertime atmospheric profile for Hamburg extracted from EAM and the fire parameters for a fuel loading of 16 g cm<sup>-2</sup>. Manins (1985) cites observations of the Hamburg plume height from 9–12 km, which is about the height of the climatological tropopause for Hamburg in July (Wilcox et al., 2012). Our simulated plume rose to 9–13 km, in close agreement with those observations. These two comparisons to observed fires indicate that the setup of our fire model produces reasonable plume heights. Badlan et al. (2017) completed a sensitivity study to grid resolution using the WRF model with horizontal resolutions of 50, 100, and 150 m for similar fire simulations and noted little difference in the overall plume, therefore we do not investigate resolution in this work. Additional simulation results are presented in section 3 examining the sensitivity of the plume height to wind speed, atmospheric moisture, and fuel loading.

## 2.2. Climate Model and Simulations

This study employs Version 1 of the E3SM Atmosphere Model (EAMv1, Rasch et al., 2019), an atmospheric general circulation model that is one component of the E3SM coupled system. EAMv1 uses a spectral finite element dynamical core on a cubed sphere grid with 72 vertical levels and is configured here with approximately 100 km horizontal resolution at the equator. The model top is 0.1 hPa, which is on average 64 km but varies locally, especially in response to thermal changes in the atmosphere due to a smoke perturbation.

EAMv1 began as a branch of the Community Atmosphere Model, Version 5.3 (CAM5) (Neale et al., 2010) and subsequent development included many major changes. Developments relevant to our application include the following: both enhanced vertical resolution and a higher model top in comparison to CAM,

linearized ozone photochemistry in the stratosphere (Hsu & Prather, 2009), an improved modal aerosol scheme for more realistic evolution of atmospheric BC (section 2.2.1), a new scheme for aerosol wet removal and transport by convective clouds (Wang et al., 2013), and a new cloud microphysics scheme (MG2) that has aerosol-cloud interactions for shallow convective and stratiform clouds (Gettelman et al., 2015).

EAMv1 is run with the active atmosphere and land component and a prescribed present-day (1981–2010 climatology) seasonal cycle for sea surface temperatures (SST) and sea ice concentrations from the Hadley-National Oceanic and Atmospheric Administration (NOAA) merged data set (Hurrell et al., 2008). The prescribed climatological SST and sea ice configuration allows for more efficient simulations that resolve the radiative forcing by the injected BC, but not the changes to SST and ocean dynamics that would occur in response. Mills et al. (2014) note that initial temperature anomalies for SOCOL3, GISS ModelE, and WACCM4 are all proportional to the shortwave radiation anomaly at the surface. Therefore atmospheric BC surface radiation deficits generated from the fixed-SST configuration in EAM are also likely proxies for the initial surface temperature response, but a dynamic ocean would be necessary to resolve the transient climate response and recovery. The emissions that are unrelated to the nuclear exchange are prescribed to near present-day (Year 2000) decadal monthly means, and the CO<sub>2</sub> volume mixing ratio is 367 ppm.

GCMs and Earth system models that have previously simulated the climate response to a regional nuclear exchange (and references to those simulations) are the NASA Goddard Institute for Space Studies ModelE, that is, GISS ModelE (Robock et al., 2007), WACCM Versions 3 and 4 with Community Aerosol and Radiation Model for Atmospheres (CARMA) aerosol (Mills et al., 2008, 2014; Reisner et al., 2018), the third generation chemistry-climate model SOCOL (i.e., modeling tools for studies of Solar Climate Ozone Links Stenke et al., 2013), and NorESM1-M (Pausata et al., 2016). Simulations have been run with prescribed SST (Mills et al., 2008), coupled to a mixed layer ocean (Stenke et al., 2013), and coupled to a dynamic ocean (Mills et al., 2014; Pausata et al., 2016; Robock et al., 2007). Model tops range from 3 hPa (approximately 40–45 km) in NorESM1-M to 5.1e–6 hPa (approximately 140 km) in WACCM4 (Mills et al., 2014). Mills et al. (2008) found BC particles reach a maximum altitude of about 0.01 hPa or 80 km, above which air pressure is insufficient to keep the particles suspended. The 0.01 hPa pressure above which settling overcomes lofting in WACCM is equivalent to the upper interface of the top level of EAM, suggesting the EAM model top should not greatly inhibit BC lofting, particularly given that the aerosol representation in EAM may favor faster settling than WACCM (see section 2.2.1).

### 2.2.1. Black Carbon Aerosol and Chemistry in EAM

The four-mode Modal Aerosol Module (MAM4 Liu et al., 2016) predicts aerosol mass mixing ratio and number concentrations in each mode: accumulation, Aitken, coarse, and primary. Species are internally mixed within modes, and particle optical properties are computed from the volume-average of the properties of the species it contains (i.e., there is no “lensing” effect). Particles are assumed to be spherical. Modes obey a lognormal size distribution with predicted geometric mean diameters and a fixed geometric standard deviation. The geometric mean is allowed to vary between upper and lower bounds, which, if encountered, result in the creation or destruction of additional particles.

In MAM4, the two carbonaceous aerosol species are particulate organic matter (POM), which scatters short-wave radiation, and black carbon (BC), which absorbs it. We use the terminology from Bond et al. (2013) to describe the composition of smoke and aerosol species, which is not always consistent across the literature. Smoke from real fires contains BC, primary organic aerosol (POA), gaseous precursors to secondary organic aerosol (SOA), and SOA itself (Bond et al., 2013). BC is pure carbon and relatively chemically inert, whereas POA and SOA (collectively OA) are particles containing carbon and other elements. Organic carbon (OC) is the carbon mass within OA. OA and BC are referred to collectively as carbonaceous aerosol.

POM as defined in MAM4 contains both organic carbon (OC) and associated other elements, analogous to OA as defined above. The mass of other elements in POM is accounted for by multiplying emissions estimates of OC by a factor of 1.4 to calculate emissions of POM. Pausata et al. (2016) is the only other regional nuclear exchange paper to include POM co-emitted with BC. The other regional nuclear exchange papers neglect emissions other than BC under the assumption that OA would be destroyed by photochemical reactions in the stratosphere, but analysis of a 2017 British Columbia wildfire smoke plume suggests that the *e*-folding chemical lifetime of OA in the lower stratosphere is 100–150 days (Yu et al., 2019), which is long enough to cause a sustained radiative forcing. Therefore, we co-emit POM with BC in some experiments (labeled “+POM”), even though EAM does not include the mechanisms for its photochemical destruction.

In the “+POM” experiments the BC:POM ratio in smoke is 1:3 for consistency with Pausata et al. (2016), who estimated this ratio for fires confined to urban areas.

The MAM4 primary aerosol mode contains only carbonaceous aerosols (BC, POM), whereas the “aged” or “accumulation” mode in MAM4 contains POM, BC, and other non-carbonaceous species. Primary and aged carbonaceous aerosols in MAM4 differ physically and chemically in important ways. Primary carbonaceous aerosol is generally smaller and hydrophobic; aged carbonaceous aerosol is generally larger and hydrophilic. The transfer or “aging” of carbonaceous aerosols from the primary to the accumulation mode occurs after particles condense eight monolayers of (hydrophilic) sulfate, particles condense enough secondary organic aerosol (SOA) to change the volume-weighted hygroscopicity by the same amount as eight layers of sulfate, or particles coagulate with hydrophilic Aitken-mode particles. Additionally, carbonaceous aerosols can also enter the “coarse” mode when they are released from evaporated raindrops, but that pathway is not further explored in this work. Coagulation in MAM4 allows for particle growth, which increases fall velocity. Size (and optical properties) were held fixed in the regional nuclear exchange studies using CARMA (Mills et al., 2014; Reisner et al., 2018). However, CARMA represents BC particles as a fractal pattern, which may be more realistic than the simplifying spherical assumption in MAM4 (Mills et al., 2014).

In addition to incorporating BC aging, parameterized wet deposition has been a focus of EAM development. EAMv1 uses a new unified aerosol convective vertical transport and scavenging parameterization (Wang et al., 2020), treating these processes simultaneously instead of sequentially. Also new is aerosol activation within updrafts above the cloud base instead of only at the cloud base. Cloud-borne aerosols may be detrained from updrafts and resuspended in the coarse mode. Below-cloud wet removal (impaction, Brownian diffusion) is unchanged.

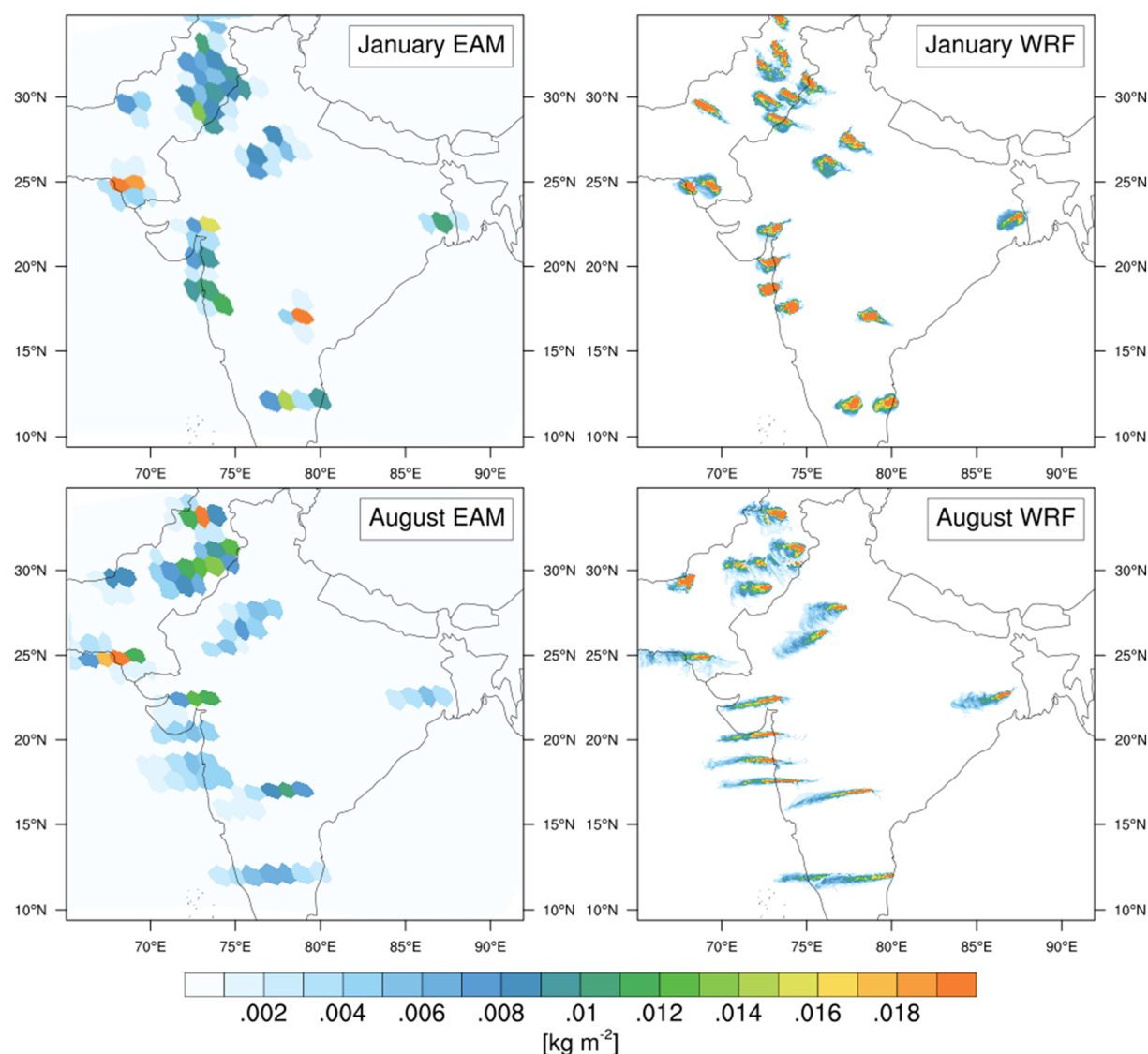
These changes decreased the BC removal efficiency in the troposphere by design. Wang et al. (2013) found that CAM5 was too readily removing BC in stratiform clouds in mid- and high latitudes by nucleation scavenging, resulting in too little BC reaching the Arctic. One cause of the overestimation of scavenging was an inconsistency in the way that the droplet nucleation code interfaced with the subgrid cloud fraction for liquid-containing clouds. This inconsistency caused the nucleation scheme to overestimate the liquid cloud fraction and therefore caused excessive wet removal. Another problem in CAM5 was overestimation of liquid clouds in cold, dry air, again causing an overestimation of nucleation scavenging. We are unaware of whether these factors leading to overestimation of wet deposition are present in CAM4 or CAM4-related models (NorESM1-M, WACCM4). Use of the new unified aerosol transport and removal scheme decreased wet deposition rates and increased the burdens of BC in remote Arctic regions by a factor of 10 in winter.

The linearized ozone stratospheric photochemistry in EAM (LINOZv2; Hsu & Prather, 2009) calculates net ozone production from perturbations to local temperature, local ozone, and ozone in the column above as independent variables. The coefficients for each independent variable come from a linearization of the modeled ozone response to small perturbations of the independent variables (e.g., changing local temperature by 4 K) in a photochemistry box model (Hsu & Prather, 2009). Our simulations contain much larger perturbations to the independent variables than were used to generate the coefficients (see section 4), but no modifications to LINOZv2 were made for this application.

### 2.2.2. Perturbing EAM Using WRF Smoke

Smoke plumes from the WRF simulations are injected into EAM at 09:30 UTC with five plumes located at each of the 20 detonation sites. This setup is shown in the top panels of Figure 2 for the January and August simulations. Interpolation from the WRF to the EAM grid is accomplished using a method that preserves the BC vertical profile (BC mass per meter) and horizontal extent. Every WRF grid cell is mapped to an EAM grid cell based on the horizontal coordinates of the WRF grid cell center and its geopotential height. Multiple WRF grid cells map to individual EAM grid cells, so after the mapping is finished, the total BC mass is summed within each EAM grid cell to obtain mass mixing ratio, which is added to the EAM initial condition file (shown in the bottom panels of Figure 2). BC mass is conserved during the mapping but mass mixing ratio is not because of differences in temperature and air density that develop as the WRF and EAM simulations diverge during the fire simulations. Temperature, pressure, moisture, and other fields affected by the large fires in WRF are not passed into EAM.

In actual fires, BC particles are formed in flames, and are externally mixed with other particles. Over a period of hours to days, BC becomes internally mixed through condensation and coagulation (Bond et al., 2013). EAM simulations begin after carbonaceous aerosols have been in the atmosphere for up to 4.5 h, the duration



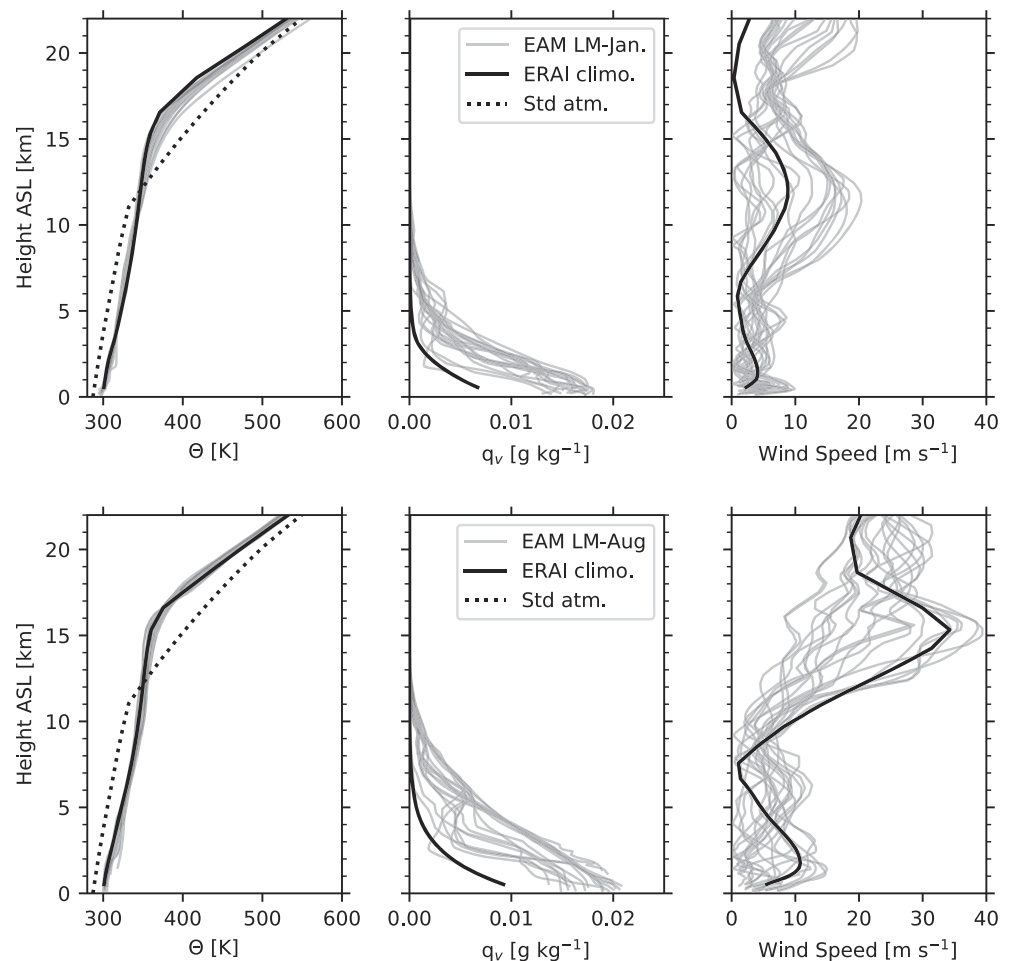
**Figure 2.** Vertically integrated BC at EAM initialization (left) derived from WRF LM simulations for 100 total fires across 20 sites (right). (top) January; (bottom) August.

of the WRF simulations, rising with a buoyant smoke plume. The physical and chemical changes undergone by carbonaceous aerosols are not included in WRF, so we must choose whether to consider them “primary” (primary mode) or “aged” (accumulation mode) when injecting into MAM4. Rather than attempt to partition BC between modes in MAM4, we inject all the mass into one mode or the other, with the intention of bounding the uncertainty due to choice of mode.

In response to the injection of carbonaceous aerosol mass, MAM4 adjusts the aerosol number in each mode such that the mode mean diameter size limits are not exceeded. In grid cells where mass is injected, the geometric mean of the number distribution of the diameter,  $d_g$ , reaches the threshold for the mode (0.10 and  $0.44 \mu\text{m}$  for primary and accumulation modes, respectively). For comparison, Mills et al. (2014) carried out experiments in CESM-WACCM using aerosols with fixed sizes of either 0.1 or  $0.2 \mu\text{m}$  diameter. Therefore,  $d_g$  for primary injections in EAM is equivalent to the smaller-particle experiment in CESM-WACCM, and  $d_g$  for accumulation mode injections in EAM is 2.2 times larger than the larger-particle experiment in CESM-WACCM.

Solar heating of the injected BC in EAM causes rapid ascent of air parcels, which necessitates reducing the model time step, in particular the vertical remapping sub-time step, to maintain numerical stability.





**Figure 3.** Soundings used to initialize WRF simulations in January (top) and August (bottom). A sounding from the ERA-Interim January climatology (averaged from 1989–2005) over south central India is used to initialize the WRF model for sensitivity tests. A sounding from EAM of instantaneous conditions at each detonation site (gray) is used to initialize the WRF model for the local meteorology (LM) simulations. Profiles are shown for potential temperature ( $\theta$ ), water vapor mixing ratio ( $q_v$ ), and wind speed. Potential temperature for the 1976 U.S. Standard Atmosphere is included for comparison and is by definition dry.

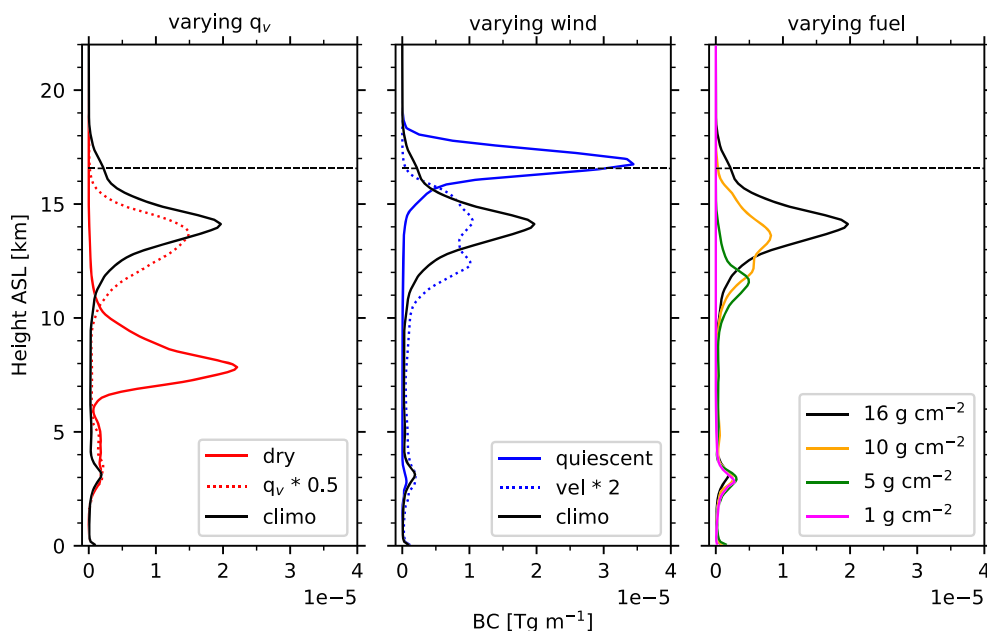
We halve it (from 15 to 7.5 min) for the first 30 months by calling the model dynamics four times, rather than twice, per physics time step.

### 3. Results From WRF Simulations of Fire Plumes

#### 3.1. Sensitivity of Fire Simulations to Wind, Moisture, and Fuel Loading

The height of a smoke plume is known to depend on both the fire parameters and local atmospheric conditions. The heat and moisture fluxes and size and shape of the fire contribute to plume rise dynamics, as do the atmospheric stability, ambient wind speed and shear, and latent heat released from condensing water vapor (Paugam et al., 2016). Larger fire size and higher heat fluxes have been shown to increase plume height (Badlan et al., 2017; Freitas et al., 2007; Penner et al., 1986), as have areal fires (i.e., circles or squares) in comparison to linear fires covering the same area (Badlan et al., 2019) through reduced entrainment of ambient air into the convective core. Higher ambient wind speeds and shear can enhance entrainment and decrease plume height (Freitas et al., 2010).

Here, we investigate the sensitivity of predicted plume height to atmospheric moisture, wind speed and fuel loading using the WRF model. In this set of fire simulations, a sounding of the climatological mean January conditions for south central India derived from the ERA-Interim reanalysis, as shown by the solid black line in Figure 3 (top panels), is used to initialize the WRF model. Notably, the tropopause height in the

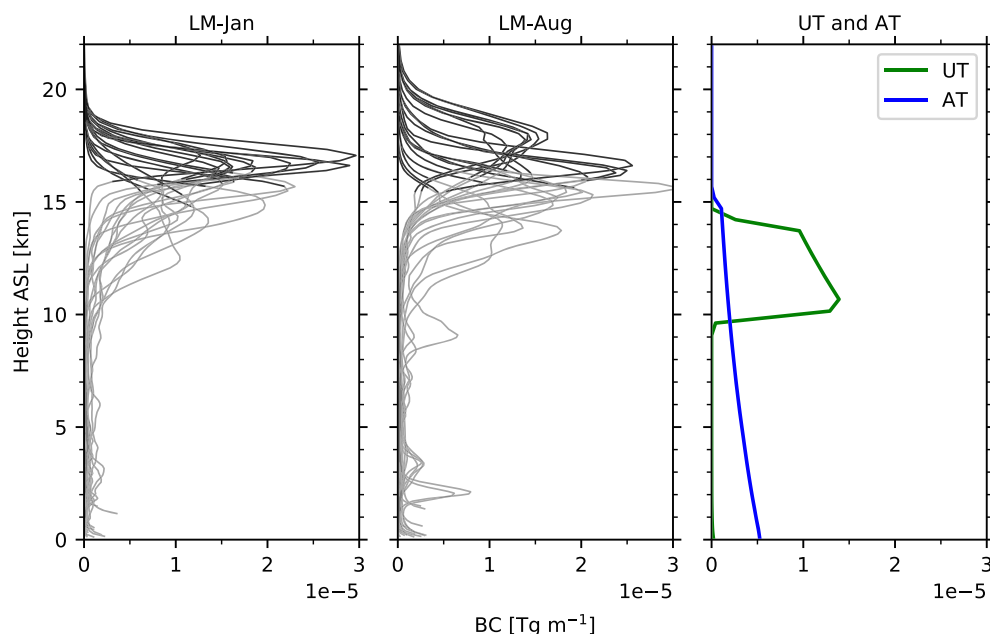


**Figure 4.** Vertical profiles of horizontally integrated BC mass from WRF fire plume simulations showing sensitivity to atmospheric moisture (left), wind speed (center), and fuel loading (right). Dashed black line is the climatological tropopause for January computed from ERA-Interim reanalysis (WMO T-lapse rate tropopause definition).

climatological sounding is much higher ( $\sim 16$  km) than in the U.S. Standard Atmosphere (11 km), which can allow for higher plumes than in simulations using a standard atmosphere. Additionally, the climatological sounding is moist, includes wind speeds of nearly  $9 \text{ m s}^{-1}$  and large directional shear (which can be seen in Figure 1). EAM produces a moister atmosphere at the injection sites on the date of detonation (Figure 3, gray lines), and in climatology (not shown). In comparison to the ERA-Interim climatology in northern India, the south Indian climatology has a higher tropopause and lower wind speeds (not shown). The south Indian region was chosen, partly, because the lower wind speeds allowed for a smaller computational domain, allowing the sensitivity studies to be completed at a reasonable computational cost.

Vertical profiles of horizontally integrated BC from the outer domain our two-domain WRF simulations are shown in Figure 4. In the first panel, atmospheric water vapor in the input sounding is varied between that of the climatological sounding, half of the climatological values, and dry. In all cases, fire parameters take the default values given in Table 1 (i.e., those for a fuel loading of  $16 \text{ g cm}^{-2}$ ), with the exception of the dry case where the moisture flux at the fire source is eliminated. Smoke emissions concentrate just below the tropopause in the moist cases, between 12 and 16 km, while in the dry case emissions are approximately 6 km lower. Similar differences in plume height between dry and moist cases were reported in Cunningham and Reeder (2009). In the middle panel, wind speed is varied between quiescent, that of the climatological sounding, and twice the speed of the climatological sounding. With a quiescent atmosphere, smoke emissions peak just above the tropopause height, injecting over 50% of the smoke emissions into the stratosphere. Plume heights decrease with increasing wind speed, as expected, though the emissions remain concentrated just below the tropopause.

Fuel loading is additionally varied between 1 and  $16 \text{ g cm}^{-2}$ , resulting in the fire source term parameters provided in Table 1. Variations in fuel loading affect the surface heat flux and total emissions of smoke and water vapor. Therefore, while total smoke emissions are 5 Tg (for 100 fires) for the cases with varied moisture and wind, emissions vary with fuel loading. Vertical profiles of horizontally integrated BC are shown for cases with varied fuel loading in the right panel of Figure 4. Reductions in fuel loading decrease the injection height. When a fuel loading of  $1 \text{ g cm}^{-2}$  is used, smoke emissions remain concentrated below 4 km, while in the cases with higher fuel loading emissions rise to above 10 km, with the majority of the smoke remaining below the tropopause.



**Figure 5.** (left and center) Vertical profiles of horizontally integrated black carbon mass for each of the 20 local meteorology (LM) WRF simulations of urban fires in January (left) and August (center). The transition between gray and black marks the local tropopause. (right) Idealized upper tropospheric (UT) (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013) and all-tropospheric (AT) (Reisner et al., 2018) smoke sources for comparison (normalized per fire for 100 fires).

### 3.2. Fire Simulations With Local Meteorology

WRF simulations of individual urban areas are initialized with local meteorology (LM) from instantaneous atmospheric soundings. The soundings are extracted from EAM in January (LM-Jan) and August (LM-Aug). Soundings for the LM-Jan case are plotted in gray in Figure 3. A fuel loading of  $16 \text{ g cm}^{-2}$  is assumed for LM simulations. The resulting vertical profiles of horizontally integrated BC mass are shown in Figure 5. The LM simulations have generally higher injection heights than simulations initialized from climatology: Almost  $2 \text{ Tg}$  of BC (40%) is directly injected above the tropopause for January and August. Although some soot from every fire is directly injected into the stratosphere, the amount that is injected into the stratosphere varies significantly from site to site due to the variable local meteorology. The local meteorology at detonation sites alters the peak BC injection height by about  $5 \text{ km}$  across the domain on a given day.

The WRF simulation initialized using the ERA-Interim climatological mean sounding from January in southern India also resulted in a partial stratospheric injection (solid black line in Figure 4), however, its peak BC mass is lower in altitude than most of the LM-Jan profiles. We attribute the high LM injection heights, both for January and August, to a combination of an EAM climatological moisture bias and the use of an arbitrary injection date which happened to exacerbate the bias. For the January injection, the instantaneous profiles show tropical-like conditions over some extratropical sites (Figure 3).

The high LM injection heights occur in combination with strong updrafts: Resolved updrafts on the  $200 \text{ m}$  inner grid can exceed  $150 \text{ m s}^{-1}$  in the upper troposphere, and updrafts exceeding  $100 \text{ m s}^{-1}$  occur as low as  $6.75 \text{ km}$ . Updraft velocities are similar to those in Reisner et al. (2019), where a different atmospheric model was used, and velocities of  $125$  to  $180 \text{ m s}^{-1}$  are reported. Upper tropospheric winds steer the majority of the BC mass within the plumes (Figure 2). The  $40 \text{ m s}^{-1}$  upper tropospheric horizontal winds in the August simulation are insufficient to prevent injection into the stratosphere. Note that the fire surface heat flux is prescribed, so surface winds cannot feedback on the fire spread or intensity.

The WRF BC vertical profiles, when summed horizontally over each detonation site, are frequently higher than have previously been used as a smoke source term for GCMs in previous studies. Approximate reconstructions of the upper tropospheric (UT) smoke source term from  $300$ – $150 \text{ hPa}$  used by Robock et al. (2007), Mills et al. (2008, 2014), Stenke et al. (2013), and Pausata et al. (2016) and an all-tropospheric (AT) injection source term approximating that used by Reisner et al. (2018) are shown in Figure 5.

**Table 2**  
*Climate Simulations Performed in EAM*

| Name                   | Smoke source           | Aerosol mass (Tg) | Mode | Month    |
|------------------------|------------------------|-------------------|------|----------|
| Upper troposphere (UT) | Idealized <sup>a</sup> | 5.0 BC            | P, A | Jan, Aug |
| All troposphere (AT)   | Idealized <sup>b</sup> | 3.8 BC            | P, A | Jan, Aug |
| Local meteorology (LM) | WRF-LM                 | 5.0 BC            | P, A | Jan, Aug |
| LM + POM               | WRF-LM                 | 5.0 BC, 15.0 POM  | P    | Jan      |
| 1 g cm <sup>-2</sup>   | WRF-climo              | 0.31 BC           | P    | Jan      |
| 5 g cm <sup>-2</sup>   | WRF-climo              | 1.56 BC           | P    | Jan      |
| 10 g cm <sup>-2</sup>  | WRF-climo              | 3.13 BC           | P    | Jan      |
| 16 g cm <sup>-2</sup>  | WRF-climo              | 5.0 BC            | P    | Jan      |

Note. “P” and “A” refer to “primary” and “accumulation,” respectively, indicating the MAM4 aerosol mode into which all of the BC is injected.

<sup>a</sup>Approximate source term used by Robock et al. (2007), Mills et al. (2008, 2014), Stenke et al. (2013), and Pausata et al. (2016). <sup>b</sup>Approximate source term used by Reisner et al. (2018).

## 4. Results From Climate Simulations

EAM simulations are completed with BC initializations from our WRF plumes (LM) and with upper troposphere (UT) and all troposphere (AT) distributions (see Figure 5). The latter two source terms enable comparison between EAM and other GCMs. All GCMs have run UT source terms (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013), whereas the AT source term is an approximation of the source term used by Reisner et al. (2018) for comparison to that study. For each BC source term (UT, AT, and LM), we conducted results for injections into the primary and accumulation modes in MAM4 in the months of January and August. Climate simulations are summarized in Table 2. In addition to the three BC source terms, climate simulations with POM and using fire emissions from different fuel loadings were also conducted (also included in Table 2).

Smoke plumes injected into EAM persist in the atmosphere for years. Long-term (lasting up to 3 to 5 yr) distribution and climate impacts of BC are discussed in section 4.1, followed by a closer examination of the initial days to weeks after the injection, because of the outsized importance of this initial time period on the eventual long-term climate impacts (section 4.2).

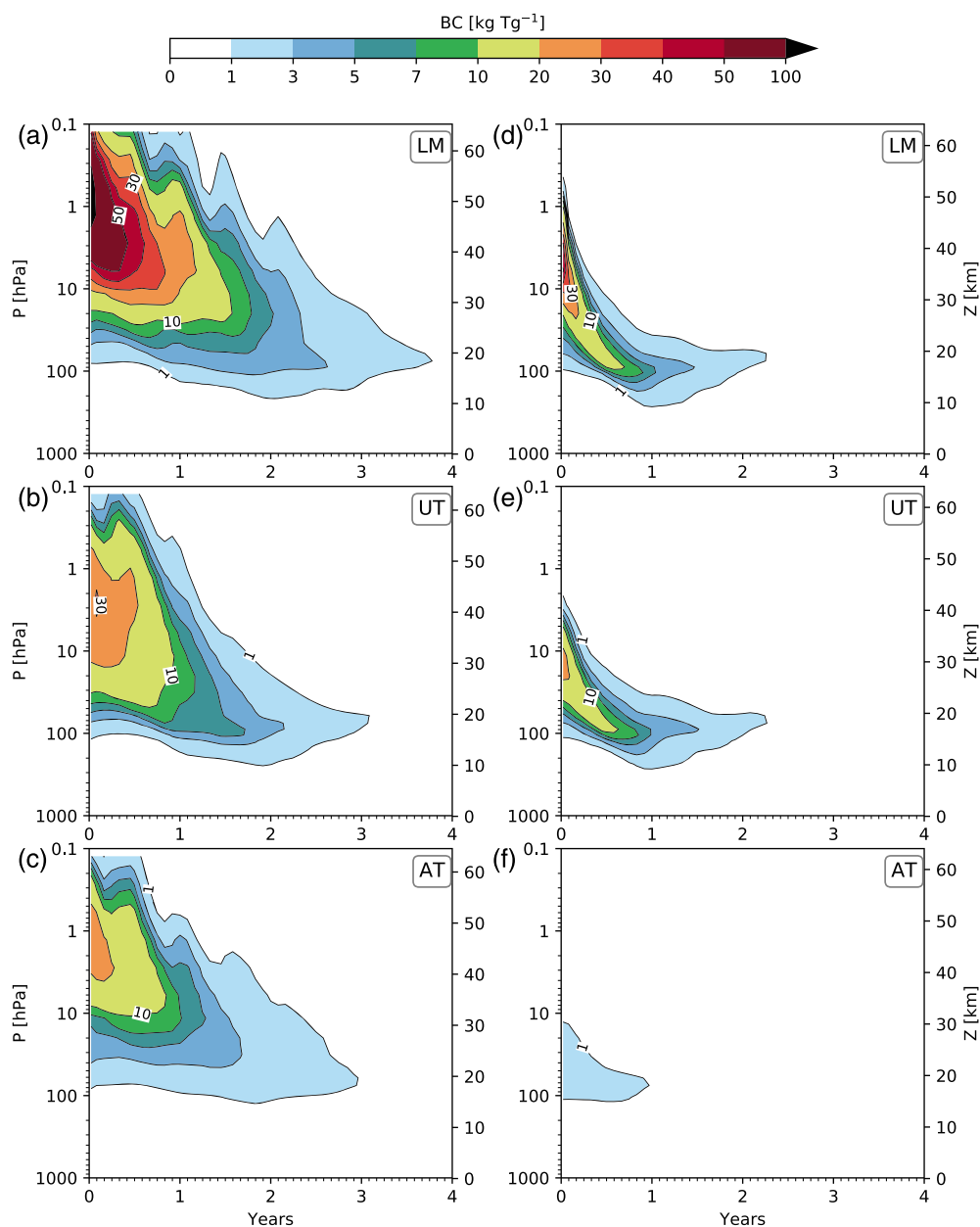
### 4.1. Long-Term Climate Impacts

#### 4.1.1. BC Global Transport and Lifetime

Area-averaged BC mass mixing ratio is shown in Figure 6 for BC injections into the primary and accumulation aerosol modes. In all simulations, BC absorbs solar radiation, heats, and radiates to the surrounding air, inducing buoyancy and self-lofting of the plume in EAM. The injected BC that is not removed by rapid wet or dry deposition in the initial few months (section 4.2) self-lofts into the stratosphere and mesosphere. The higher initial injection heights of the LM case favor higher self-lofting, as is evident by comparing results using the LM injection to those using the UT and AT injections (Figure 6). Area-averaged BC mass mixing ratios in Figure 6 are plotted for comparison to CESM-WACCM in Figure 1 of Mills et al. (2014) and Figure 9 of Reisner et al. (2018).

Injected particle size and hygroscopicity are important factors affecting BC lofting, spread, and lifetime. These parameters are adjusted by injecting all the carbonaceous aerosol mass into either the primary carbon or accumulation aerosol mode of MAM4. For all injections, injecting in the accumulation mode (which increases the mean particle diameter and makes the aerosols initially hydrophilic) decreases the peak altitude, lifetime, and climate impacts of BC.

As a complement to plotting the mixing ratio of BC (Figure 6), and for consistency with the vertical profiles shown in Figures 4 and 5, the BC mass profile (Tg m<sup>-1</sup>) in EAM is plotted in Figure 7. The relatively reduced mass of the 3.8 Tg AT primary injection (Figure 7c) is clearly distinguishable from the 5.0 Tg UT primary injection (Figure 7b), whereas the mixing ratios of these experiments are similar (Figures 6b and 6c) due to differences in lofting height. The absolute mass profile also shows that most BC mass does not loft above the



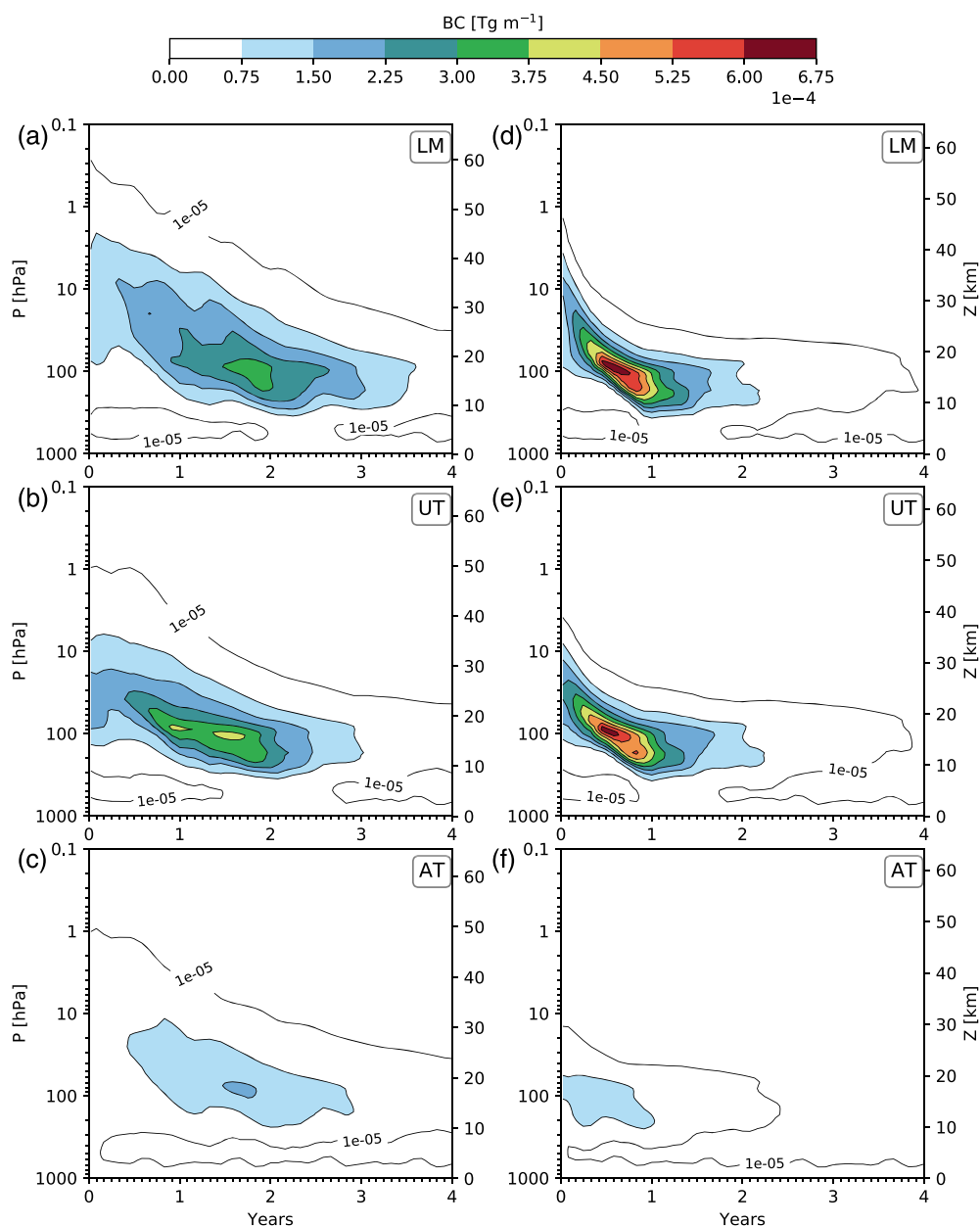
**Figure 6.** Area-averaged BC mass mixing ratio versus time since injection in EAMv1 for January injections into the primary mode (a–c) and accumulation mode (d–f). LM, UT, and AT describe the source term. Colors and contours for comparison to Figure 1 in Mills et al. (2014) and Figure 9 in Reisner et al. (2018).

relatively dense air of the lower stratosphere (e.g.,  $\sim 100$  hPa), even when peak mass mixing ratios are found as high as  $\sim 0.1$ – $1$  hPa.

Because BC mass is concentrated in the lower stratosphere, lower stratospheric winds are important to its global transport. Vertically integrated BC for selected months of the first year after a UT-Jan primary injection is shown in Figure 8. The poleward transport is accomplished via the Brewer-Dobson circulation (BDC). A comparison of Figure 8 to CESM-WACCM as shown in Figure 11 of Reisner et al. (2018) (plotted on similar color scales and contour intervals) shows that the timing of BC's zonal and meridional transport to remote regions is similar, although transport over the south pole is slower in the EAM simulation.

BC burden in the atmosphere is shown in Figure 9 for EAM simulations with UT, AT, and LM January injections. Primary mode injections are shown as solid lines, connected by a shaded area to the corresponding

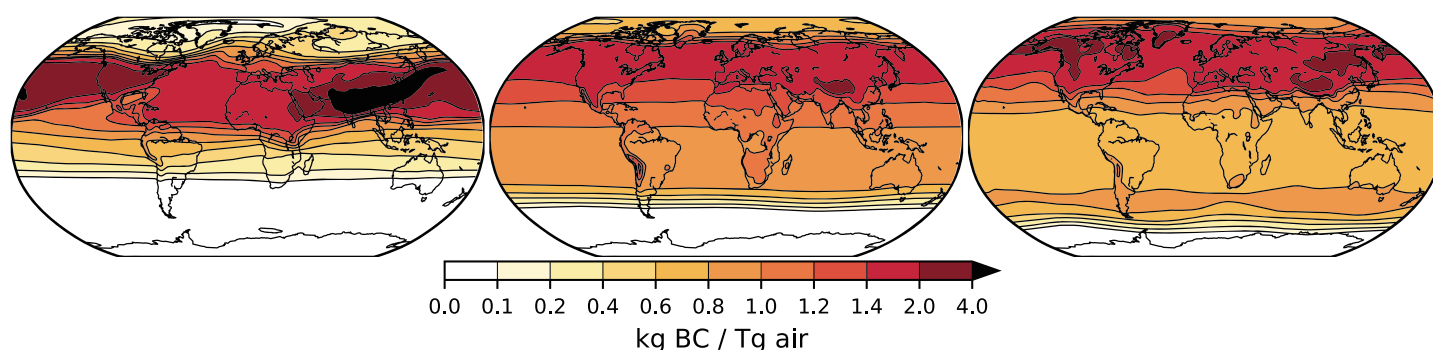




**Figure 7.** Area-integrated BC mass profile versus time since injection in EAMv1 for January injections into the primary mode (a–c) and accumulation mode (d–f). LM, UT, and AT describe the source term.

accumulation mode injection. Also shown is the EAM August simulation, and for comparison, BC burden from previous studies using WACCM (Mills et al., 2014; Reisner et al., 2018). For 5 Tg injections in EAM, a similar mass of BC remains in the atmosphere for the first year of the simulation, regardless of whether the initialization is LM, UT, primary, or accumulation mode. Subsequent BC removal occurs earlier when it is injected into the accumulation mode, as the larger particles settle back into the troposphere faster than particles in the primary mode. EAM August simulation results are similar to those from January, with slightly higher lofting of BC, likely due to greater insolation, and thus a longer atmospheric lifetime. When BC mass is included in the lower troposphere, as in the AT injection, several Tg of BC mass are removed within the first months, and more for accumulation mode injections than primary injections.

The *e*-folding lifetime of BC mass in the stratosphere ranges from 1.1 yr for the AT smoke injected into the accumulation mode to 3.0 yr for the LM plumes injected into the primary mode in August (Figure 9). UT smoke injected into the primary mode for January simulations in EAM is has an *e*-folding lifetime of

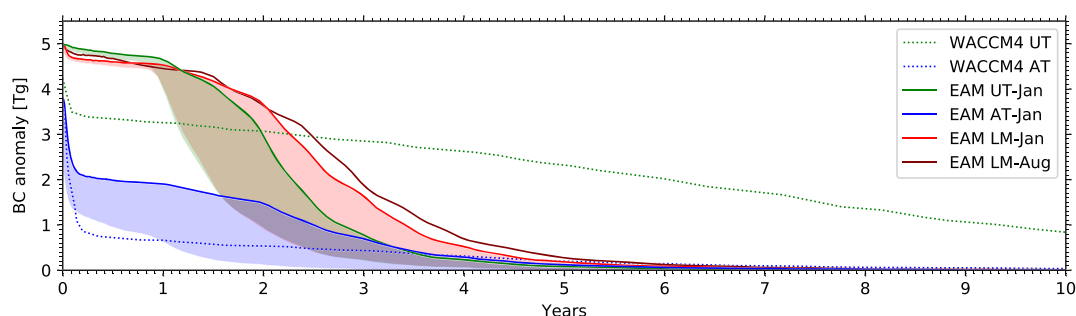


**Figure 8.** Vertically integrated BC mass mixing ratio anomaly in EAM for an upper tropospheric (UT) 5 Tg January injection into the primary mode, shown for Months 1 (left), 6 (center), and 12 (right) after detonation. For comparison with CESM-WACCM, see Figure 11 in Reisner et al. (2018)

about 2.3 yr, which is 74% shorter than the 8.7 yr *e*-folding lifetime of UT smoke in CESM-WACCM4 with CARMA aerosols (Figure 9), and shorter than other GCMs for January UT injections (see Table 3 for a comprehensive comparison of BC lifetime in previous studies). For an accumulation mode UT injection in EAM, the *e*-folding lifetime is only 1.4 yr.

Several factors likely contribute to the decreased BC lifetime in EAM relative to other climate models for the same initial 5 Tg BC UT smoke initialization. The lower model top in EAM may inhibit lofting. However, the EAM model top is higher than NorESM1-M, but EAM simulates a shorter BC stratospheric lifetime (Table 3). Stratospheric dynamics may also play a role: The BDC transports BC in the lower stratosphere to the high latitudes where the most deposition occurs, so if this circulation is systematically different between models, lifetime could be affected. (Mills et al., 2008, 2014) document a slowdown of the BDC in response to BC, increasing the BC stratospheric lifetime; BDC recovery takes longer in CESM-WACCM4, which is coupled to an ocean model, relative to CESM-WACCM3, which has prescribed SST. Therefore, coupling EAM to an ocean model could also increase simulated BC stratospheric lifetime.

The most important factors affecting BC lifetime are likely related to aerosol modeling. MAM4 in EAM assumes spherical carbonaceous particles with density  $1,700 \text{ kg m}^{-3}$ ; CARMA as configured in (Mills et al., 2008, 2014) assumes a spherical particle with density  $1,000 \text{ kg m}^{-3}$  (the lower density accounts for the voids within BC particle). We tested the importance of BC density on lifetime in EAM by decreasing BC density to  $1,000 \text{ kg m}^{-3}$ , which increases BC stratospheric *e*-folding time by about 5 months for a UT, primary mode injection. Therefore, density explains some, but not all, of the difference in lifetimes between EAM and WACCM-CARMA. Furthermore, EAM includes coagulation and aging, which allows particles to increase in size and fall speed. Toon et al. (2019) configured CESM-WACCM with an updated CARMA allowing for coagulation of explicitly fractal particles among other changes and found stratospheric BC lifetimes were reduced to about 5 yr for a 5 Tg injection. Thus, it seems reasonable that treating particles as spheres and allowing for coagulation and aging could explain the even shorter EAM lifetimes. Our method of injecting aerosol mass maximizes the particle sizes (and their fall speeds) in each mode. We performed a



**Figure 9.** BC burden anomaly versus time since injection for EAMv1 simulations and comparable WACCM4 simulations. The solid lines are EAMv1 experiments with injection into the primary aerosol mode in MAM4; shaded areas fill the space between the corresponding accumulation mode injection. WACCM data are for simulations using a 100 nm fixed radius for BC (Mills et al., 2014; Reisner et al., 2018).

**Table 3**  
*BC Stratospheric e-Folding Lifetime for Models Simulating a 5 Tg Upper Tropospheric Injection*

| Model       | Lifetime (yr)    | Reference                                  |
|-------------|------------------|--------------------------------------------|
| EAMv1       | 1.4–2.3          | —                                          |
| GISS ModelE | 6                | Robock et al. (2007)                       |
| WACCM3      | 6.5 <sup>a</sup> | Mills et al. (2008)                        |
| SOCOL3      | 4.0–4.6          | Stenke et al. (2013)                       |
| WACCM4      | 8.4–8.7          | Mills et al. (2014), Reisner et al. (2018) |
| NorESM1-M   | 4                | Pausata et al. (2016)                      |

*Note.* The range for EAMv1 is defined by injections into the accumulation mode (lower bound) and primary mode (upper bound) in January.

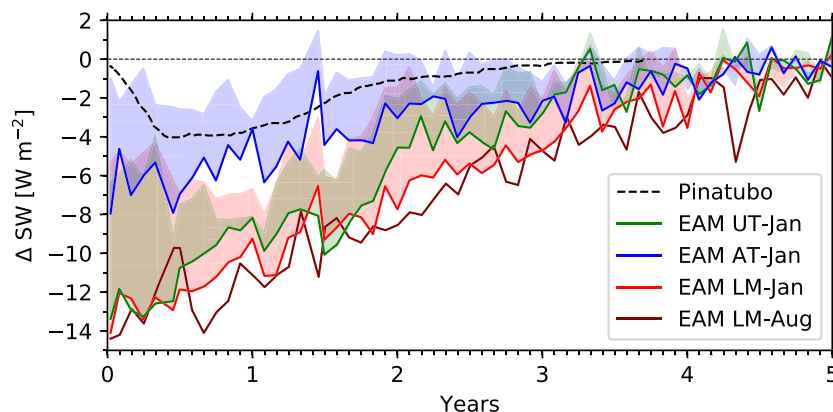
<sup>a</sup>Reported in Mills et al. (2014).

test simulation in which primary BC aerosols are injected into EAM via the emissions files over a half hour period for a UT injection using the default mean diameter ( $d_g = 0.08 \mu\text{m}$ ). Coagulation in this test plume caused rapid growth until the mean diameter reached the upper bound and lifetime was unchanged.

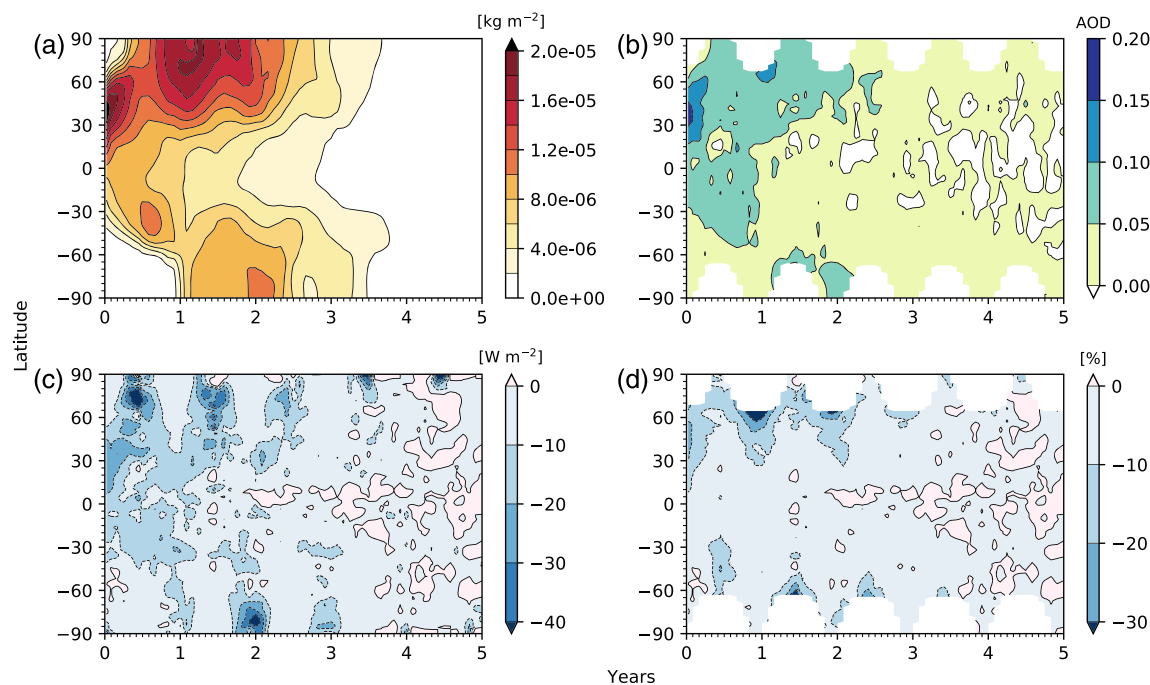
#### 4.1.2. Surface Radiation, Temperature, and Ozone

The atmospheric BC causes global-average surface solar radiation deficits which range widely according to how much BC reaches the stratosphere and whether BC is injected into the primary or accumulation mode. Accumulation mode 5 Tg injections (LM and UT) cause a shortwave radiative forcing of  $6\text{--}7 \text{ W m}^{-2}$  averaged globally over the first year, and  $11\text{--}12 \text{ W m}^{-2}$  for primary injections (Figure 10). For comparison, a result of the shortwave anomaly from a GCM simulation of the Mount Pinatubo eruption in 1991 (Oman et al., 2005) is overlaid on Figure 10. Primary mode injections cause larger perturbations than accumulation mode injections even when the BC burden is the same, in part, because primary injections have greater aerosol number per BC mass. Primary injections also result in more BC reaching the stratosphere, higher lofting, and longer BC lifetime in the stratosphere. Differences between the accumulation and primary injections are at a maximum in the initial months after injection but reduce over time as primary BC in the stratosphere ages into the accumulation mode. August simulations result in similar reductions in surface solar radiation to those in January, therefore with similar climate responses seen in both January and August, the remaining discussion focuses on the January simulations.

Shortwave radiative perturbations vary in time and space due to the transport and evolution of the BC plume, local insolation, and dynamic effects (Figure 11). Even for the relatively low insolation of the first winter and early spring, shortwave radiative deficits at the surface exceed  $20 \text{ W m}^{-2}$  from the northern tropics to  $45^\circ\text{N}$ , a reduction of 15–20% (Figures 11c and 11d). Zonal-mean shortwave deficits exceeding  $30 \text{ W m}^{-2}$  are



**Figure 10.** Monthly mean downward SW radiation anomalies at the Earth's surface versus time since injection. The solid colored lines are EAMv1 simulations with injection into the primary aerosol mode; shaded areas fill the space between the corresponding accumulation mode injection. Also shown is a simulation by Oman et al. (2005) of the June 1991 Mount Pinatubo eruption (results are extracted from Robock et al., 2007).



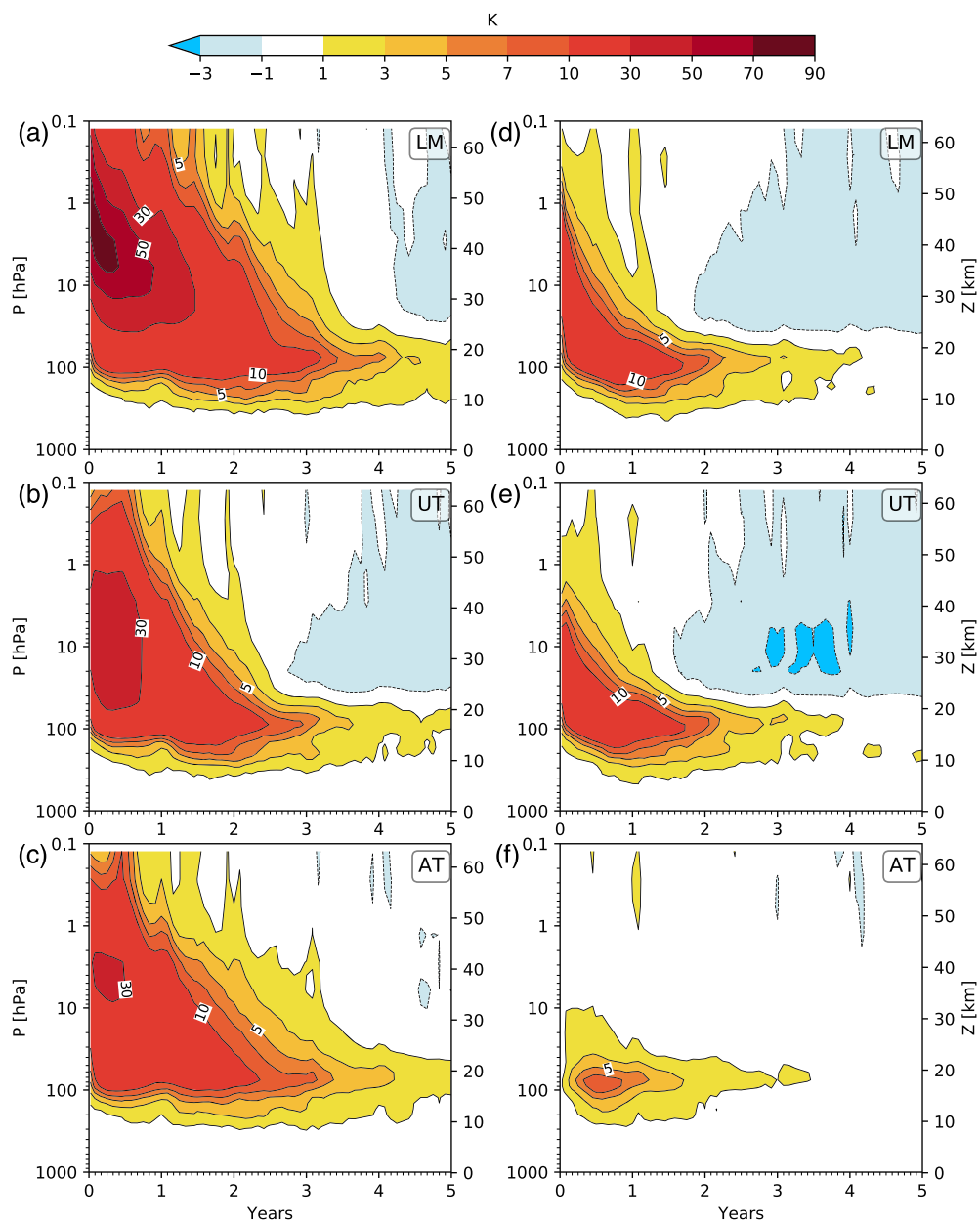
**Figure 11.** Zonal mean monthly anomalies versus time since injection for a 5 Tg BC injection into the primary mode in January (LM-Jan). (a) Vertically integrated BC; (b) aerosol optical depth in the visible band; (c) downward solar radiation at the surface ( $\text{W m}^{-2}$ ); (d) downward solar radiation at the surface as a percent of downward solar radiation in the control experiment (%).

found in the summertime Northern hemisphere high latitudes ( $60\text{--}90^\circ\text{N}$ ) in the first two summers, and in the Southern Hemisphere high latitudes for the second summer after injection.

The global radiative perturbations induced by primary mode 5 Tg injections in EAM are comparable to those simulated by other GCMs for the first 2–3 yr. For a UT-Jan 5 Tg BC injection, EAM's initial (first month)  $13 \text{ W m}^{-2}$  shortwave forcing at the surface is  $1\text{--}2 \text{ W m}^{-2}$  larger than in WACCM (Mills et al., 2014) and SOCOL (Stenke et al., 2013) (see Figure 3 in Mills et al., 2014), which may be explained by EAM's reduced rapid rainout relative to these models. The GISS ModelE, which like EAM, has little rapid rainout, simulates larger radiative forcing, approximately  $13\text{--}16 \text{ W m}^{-2}$  for the first year (Robock et al., 2007). A potential explanation for the increased forcing in GISS ModelE is lack of coagulation. These relatively small inter-model differences grow much larger after 2–3 yr, when the shorter BC lifetime in EAM results in reduced surface radiative forcing relative to other GCMs.

The 3.8 Tg AT primary injection in EAM (solid blue line in Figure 10) causes a  $5.9 \text{ W m}^{-2}$  global average shortwave radiative forcing for the first year. This sustained first-year forcing in EAM is greater than Reisner et al. (2018) find in CESM-WACCM for a comparable injection in any individual month, and far larger than their first-year average. The different forcings between GCMs when initialized with a similar AT 3.8 Tg BC source term can be explained by EAM's reduced rapid rainout for primary mode (hydrophobic) injections, which allows more BC to loft into the stratosphere relative to CESM-WACCM (Figure 9). Note that the injected BC is hydrophilic in CESM-WACCM; injecting into the accumulation mode (hydrophilic) in EAM results in comparable rapid rainout to CESM-WACCM (Figure 9), and reduced radiative forcing (Figure 10).

The use of prescribed SST precludes realistic simulation of surface temperature changes in response to the radiative forcing caused by BC. Nevertheless, surface temperatures over land during first 6 months after BC injection cool by approximately 1 K globally and by more than 2 K over the northern midlatitudes for 5 Tg January injections (UT and LM). Given that EAM primary mode injections result in similar surface radiative forcing for the first 2–3 yr as in other GCMs, a similar temperature response, with a global  $1\text{--}1.5 \text{ K}$  surface cooling by Year 3, is expected in a fully coupled model configuration for a primary mode injection. Cold anomalies would be largest at middle and high latitudes and over land. Surface temperature would recover more slowly than radiative forcing due to the thermal inertia of the ocean, likely extending beyond 5 yr. For accumulation mode injections of 5 Tg BC, the surface temperature perturbation would be less severe

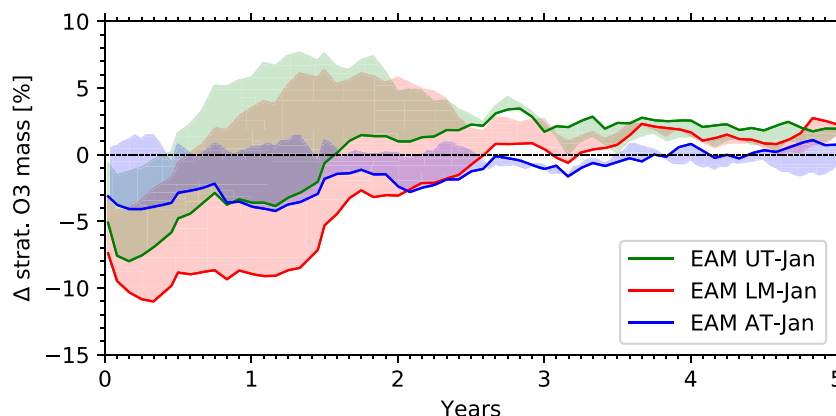


**Figure 12.** Area-averaged temperature anomaly versus time since injection in EAMv1 for January injections into the primary mode (a–c) and accumulation mode (d–f). LM, UT, and AT describe the source term.

and shorter in duration than primary mode injections, but still significant. For context, accumulation mode injections of 5 Tg of BC (UT or LM) cause approximately double the radiative forcing at the surface as was caused by the eruption of Mount Pinatubo in 1991; the forcing associated with the eruption caused a satellite-observed 0.5–0.7 K global mean lower tropospheric temperature drop (Soden et al., 2002).

The vertical profile of temperature anomalies (Figure 12) has a similar spatial pattern to the vertical profile of mass mixing ratio (Figure 6) because, although thinner air supports less BC mass, it also requires less energy to heat. Therefore, many of the results from mixing ratio translate to temperature. Global mean temperature anomalies in EAM reach a maximum of ~80 K in the upper stratosphere and mesosphere for primary LM cases, and remain over 50 K from 1–10 hPa for about 1 year (Figure 12). In contrast, the maximum heating anomalies for the AT injection into the accumulation mode are in the lower stratosphere at ~90 hPa, and are only ~9 K.





**Figure 13.** Stratospheric ozone mass anomaly versus time since injection. The solid colored lines are EAMv1 simulations with injection into the primary aerosol mode; shaded areas fill the space between the corresponding accumulation mode injection.

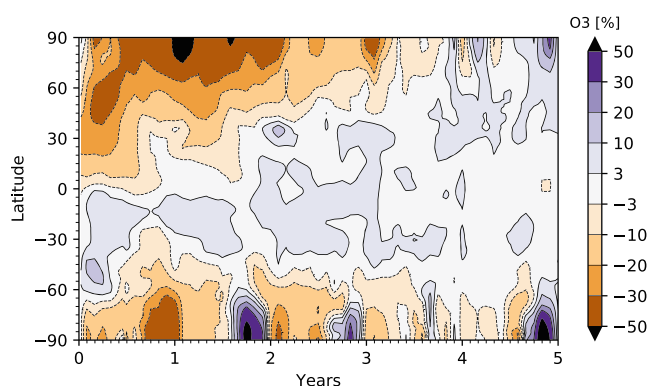
In comparison to other GCMs, the magnitude of the maximum globally averaged stratospheric temperature anomaly induced by primary 5 Tg UT injections in EAM is lower than those in CESM-WACCM, GISS ModelE, and SOCOL3 (Mills et al., 2014; Robock et al., 2007; Stenke et al., 2013), and higher than NorESM1-M (Pausata et al., 2016). The difference is likely explained by BC lofting altitude and the reduction of BC number as primary BC ages into the accumulation mode (which also tends to lower the altitude). Consistent with BC mass mixing ratio and burden, the duration of temperature anomalies is shorter in EAM relative to other GCMs, especially for the accumulation mode injections. A cooling of about 1–3 K from 20–0.1 hPa is evident in the LM and UT injections, regardless of mode, although the timing of its onset varies. Zonal-mean cooling was also simulated in CESM-WACCM4 above 1 hPa (Mills et al., 2014; Reisner et al., 2018), extending down to 10 hPa in CESM-WACCM3 (Mills et al., 2008).

Stratospheric ozone changes relative to the unforced simulation in EAM are driven by stratospheric warming and circulation changes. Global stratospheric ozone mass losses for 5 Tg primary injections peak at ~8–11% during the first year (Figure 13). Less ozone loss occurs for accumulation mode injections, likely because the larger particles heat the stratosphere less and at lower altitudes than the maximum concentration of ozone, after the first year.

Mills et al. (2014) report 20–25% ozone depletion in CESM-WACCM from the second through fifth years after detonation. The less severe ozone depletion in EAM relative to CESM-WACCM for 5 Tg injections is consistent with its reduced stratospheric warming. Additionally, Mills et al. (2014) and Stenke et al. (2013) note that photolysis of enhanced stratospheric water vapor also contributes to ozone depletion, but this chemical pathway is not fully included in our simulations. We also note that EAMv1 prescribes the tropospheric ozone to the decadal climatology and shows weaknesses in representing the interactions near the tropopause (Tang et al., 2020). These weaknesses are unlikely to change the main conclusions of the present study but will be corrected by the new ozone scheme and included in our future work.

The zonal mean spatial pattern of stratospheric ozone depletion for a primary mode 5 Tg BC injection in EAM (Figure 14) is similar to the chemistry-climate models SOCOL3 (Stenke et al., 2013) and CESM-WACCM (Mills et al., 2008, 2014), but with decreased magnitude and a faster recovery. Losses are greatest in the northern high latitudes: 30% or greater ozone reductions are sustained for 2 yr. The northern mid-latitudes lose 30% or more in the first spring and summer following the detonation, with greater than 10% losses sustained for 2 yr. The tropics and southern mid-latitudes experience little change or an increase in ozone. Mills et al. (2008) credit the initial southern mid-latitude ozone gains in CESM-WACCM to the disruption of the BDC caused by the rising BC plume; circulation changes also likely trigger the initial ozone increases in EAM.

Deposition of BC occurs in discrete phases: a rapid removal from the atmosphere in the initial hours to months after EAM initialization (section 4.2) and more gradually over months to years as BC sediments from the stratosphere into the troposphere and is removed, primarily by wet deposition. Deposition occurs globally but is at a maximum in the northern middle and high latitudes (Figure 15). Accumulation mode



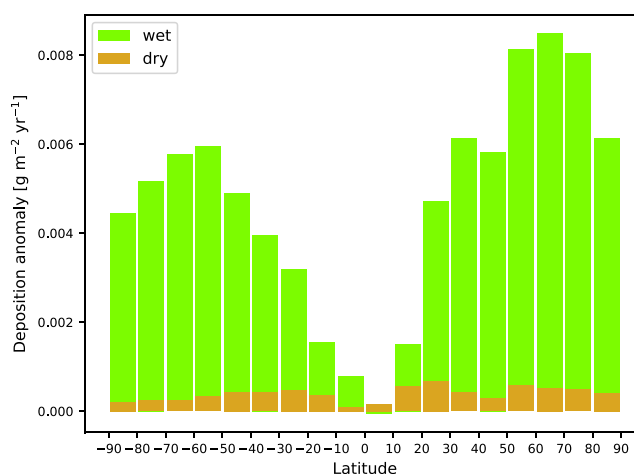
**Figure 14.** Zonal mean stratospheric ozone column mass anomaly versus time since injection for the LM-Jan primary mode 5 Tg BC injection.

injections sediment into the troposphere fastest during the first boreal winter after the injection; primary mode injections sediment fastest in the second winter (Figure 9). After settling into the troposphere, the dominant removal pathway is activation, uptake by clouds, and removal by wet deposition (precipitation) (Figure 15). Wet deposition includes impactation by falling rain, but the impactation anomalies are negligible relative to nucleation scavenging. Dry deposition of BC also contributes to removal at all latitudes.

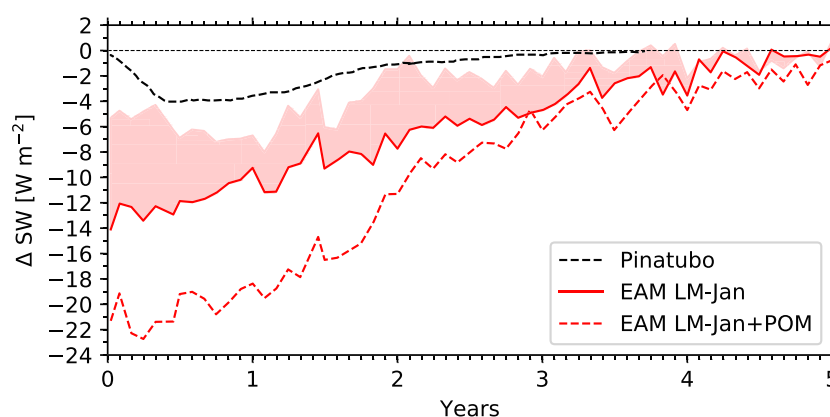
#### 4.2. Rapid Lofting and Removal

The initial hours to days after EAM initialization, while some or all of the BC is in the troposphere and potentially subject to removal, play an important role in determining the climate impacts over the subsequent months to years. Note that all modeling efforts of this scenario, including ours, assume that 20% of the BC by mass is removed during the fire, before injection into the GCM. After injection into the GCM, all GCMs simulating a 5 Tg UT injection show a rapid reduction in BC burden for the initial days to weeks, followed by a stabilization as the remaining BC escapes into the stratosphere. However, there is a wide range of this rapid deposition across GCMs: In the first 4 months, CESM-WACCM3, CESM-WACCM4, and NorESM1-M deposit 1.0, 1.6, and 1.5 Tg BC, respectively (Mills et al., 2008, 2014; Pausata et al., 2016); GISS ModelE rapid removal appears to be about 0.1–0.2 Tg (Figure 12 in Robock et al., 2007).

Rapid removal is dependent on parameterizations for dry deposition, transport, parameterized hygroscopicity (fixed or evolving) and activation, and wet removal. Assumptions about the initial hygroscopicity of BC may play an important role in some GCMs: WACCM assumes initially hydrophilic BC, whereas GISS ModelE assumes a 24 h  $e$ -folding aging timescale from hydrophobic to hydrophilic. Similar to an EAM



**Figure 15.** Deposition rate anomalies in EAM for the third year after UT-Jan primary mode injection. The third year is the year of maximum deposition for this particular injection.



**Figure 16.** Monthly mean downward shortwave radiation deficit at the surface versus time since injection for LM-Jan simulations. Shaded area fills the space between primary mode (solid line) and accumulation mode injection. POM simulation is initialized with 15 Tg POM in addition to 5 Tg BC.

primary injection, NorESM1-M assumes initially hydrophobic BC and explicitly resolves the aging process by coating BC with hydrophilic sulfate.

The EAM rapid deposition anomaly for UT smoke is only 0.1 Tg for primary mode injections and 0.2 Tg for accumulation mode injections, less than all other GCMs except GISS ModelE. Rapid deposition for the UT and LM injections is relatively insensitive to assumptions about the hygroscopicity and size of injected aerosol (Figure 9). These injections are initialized high enough that self-lofting into the stratosphere occurs before significant removal by wet deposition. The LM case has about 0.3 Tg rapid removal, but again this rapid removal is insensitive to injection mode because the removed BC is initialized beneath about 4 km (Figure 5), which is low enough that it is removed whether it is injected in the primary or accumulation mode. Because the relatively low rapid washout for UT and LM smoke is not caused by assumptions of BC hygroscopicity, it may be related to the changes made to aerosol activation and nucleation (Wang et al., 2020). In contrast, the AT injection is very sensitive to injection mode because its smoke is initialized throughout the troposphere. Rapid deposition ranges from 1.7 Tg for a primary mode injection to 2.6 Tg for an accumulation mode injection.

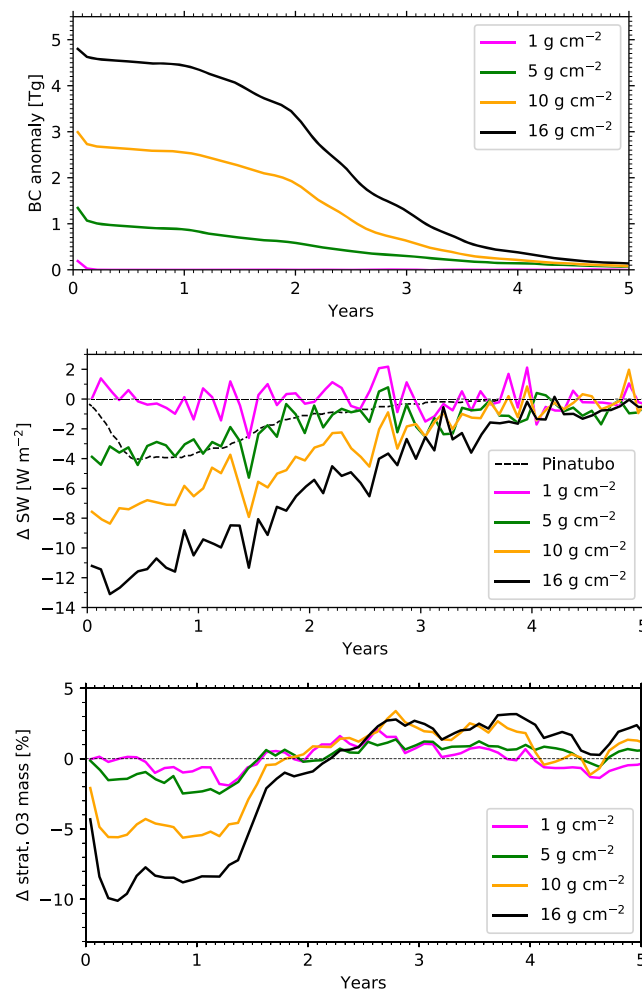
#### 4.3. Sensitivity of Climate Impacts to Primary Particulate Organic Matter

Injecting BC alone may be insufficient to simulate the climate impact of large fires. Co-emitting 15 Tg, or a 3:1 mass ratio of POM (pure OC) to BC, doubles the radiative perturbation at the surface to about 17–23 W m<sup>-2</sup> sustained for the first 2 yr by scattering solar radiation in the stratosphere (Figure 16). This deficit is larger than any models have simulated for a 5 Tg injection of BC and is also larger than Pausata et al. (2016) simulated when injecting either 15 or 45 Tg POM in NorESM1-M (their POM is about half OC by mass). However, oxidation, an important sink of POM in the stratosphere, is not currently included in EAM. Therefore, the lifetime of POM in the stratosphere, and the duration of its contribution to the radiative forcing (Figure 16), are likely overestimated in EAM. Another consideration is that the internal mixing assumption treats BC and POM as a single particle with volume-averaged optical properties. Despite these uncertainties about how to represent POM, we include it here to help bound the severity of the initial radiative perturbation.

#### 4.4. Sensitivity of Climate Impacts to Fuel Loading

Here, we investigate the climate impacts of smoke injections that differ only by the assumed fuel loading parameter. A set of WRF simulations are carried out with 1, 5, 10, or 16 g cm<sup>-2</sup> of fuel (Table 1). In all four simulations, the atmosphere is initialized from ERA-Interim climatological mean January conditions in southern India (Figure 3). The resulting vertical profiles of BC from these simulations are shown in Figure 4 (right panel). To initialize climate simulations, the WRF BC plume is replicated at the location of each detonation in EAM, accounting for local differences in surface elevation and tropopause height relative to the location of the fire simulation by scaling each vertical profile by the thickness of the troposphere. Only January injections into the primary mode are considered in this sensitivity study (Table 2).

Emissions from fire simulations with higher fuel loads cause a stronger and longer-lived radiative forcing at the surface in EAM (Figure 17). The relationship between fuel loading and radiative forcing is approximately



**Figure 17.** (top) BC burden anomaly for the fuel loading sensitivity tests. (middle) Downward shortwave radiation flux anomaly at the surface. (bottom) Percent change in global stratospheric O<sub>3</sub> mass.

linear, with the exception of the 1 g cm<sup>-2</sup> experiment because its injection height, from 2–4 km, results in rapid removal in EAM: Over 99% of the BC injected in the 1 g cm<sup>-2</sup> fuel loading simulation is removed during the first month. In contrast, a fuel loading of 5 g cm<sup>-2</sup> results in injection heights in the upper troposphere (greater than 10 km), and about 15% removal during the first month. The 5 g cm<sup>-2</sup> fuel loading scenario, with its total injection of about 1.5 Tg BC (Table 1), causes a Mount Pinatubo-like globally averaged solar radiation deficit at the surface, both in magnitude and duration. Ozone depletion is also approximately linear with fuel loading. Once in the stratosphere, BC *e*-folding lifetime for the fuel sensitivity experiments is a consistent 2.4–2.5 years in EAM (Figure 17).

## 5. Summary

We reexamine the climate and environmental consequences of a South Asian regional nuclear exchange scenario, in which 100 simultaneous mass urban fires are ignited by one hundred 15 kt nuclear detonations. Our multiscale modeling approach uses the WRF weather model and the EAM climate model, neither of which has previously been applied to this problem. We consider a range of possibilities for aerosol properties, smoke composition, and fuel loading, resulting in an improved understanding of model sensitivity to these factors.

Consistent with most previous studies of this scenario (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013), we first consider mass urban fires resulting in an emission of 5 Tg of BC. Using an assumed fuel loading of 16 g cm<sup>-2</sup>, intense fires with smoke plumes that penetrate into the

stratosphere are simulated in WRF. When injected as a source term into EAM, these plumes cause a significant global surface radiative forcing. Depending on the characteristics of the emitted BC aerosols that we tested, the range of surface shortwave radiative forcings (averaged over Year 1) ranges from about  $7 \text{ W m}^{-2}$  (accumulation mode injections) to  $12 \text{ W m}^{-2}$  (primary mode), increasing to  $21 \text{ W m}^{-2}$  with the inclusion of 15 Tg POM. This is a larger range than has been simulated in other studies using alternative GCMs. The duration of the forcings is shorter in EAM than in those GCMs (Mills et al., 2008, 2014; Pausata et al., 2016; Robock et al., 2007; Stenke et al., 2013), due to reduced BC lifetime in the stratosphere (*e*-folding time 1.4–2.3 yr for January UT injections into the accumulation and primary modes, respectively). In EAM, the primary mode injections of 5 Tg BC cause comparable radiative forcing at the surface as in the aforementioned GCMs for 2–3 yr. Although the configuration used does not enable a realistic surface temperature response to this radiative forcing, a comparison to those GCMs shows that this forcing would likely cause 1–1.5 K global cooling in fully coupled simulations with a dynamic ocean component. Accumulation mode injections would cause a less dramatic cooling, but larger than the 0.5–0.7 K caused by the 1991 eruption of Mount Pinatubo. Global ozone losses are also very sensitive to injection mode, ranging from a maximum of about 4% for accumulation mode to 11% for primary mode 5 Tg injections.

The assumed  $16 \text{ g cm}^{-2}$  fuel loading and 100% burn rate for the fire is actually uncertain, and in fact, Reisner et al. (2018) assume only  $\sim 1 \text{ g cm}^{-2}$  fuel loading. Reisner et al. (2018) points out that Indian and Pakistani cities are built of concrete, and therefore, firestorms that erupted in fuel-rich Hiroshima and Hamburg would not occur. Our simulations, using  $1 \text{ g cm}^{-2}$ , cause no global radiative forcing, because the BC emitted into the lower and middle troposphere is quickly removed by EAM. We believe that differences in assumed fuel loading partially account for the different results between Mills et al. (2014) and Reisner et al. (2018). We note, however, that the total BC mass in Reisner et al. (2018) is 3.8 Tg, while in our simulation BC mass is 0.31 Tg for the  $1 \text{ g cm}^{-2}$  fuel loading case, so that our simulation has much less BC, along with a significantly different vertical distribution of the BC. Additionally, we considered intermediate fuel loading scenarios. If the fuel loading is  $5 \text{ g cm}^{-2}$ , our simulations show that fires would generate sufficient surface heat flux to inject BC into the upper troposphere, where it self-lofts into the stratosphere. If fuel loading exceeds  $10 \text{ g cm}^{-2}$ , a fraction of BC is directly injected into the stratosphere. Using spatially varying instantaneous meteorology for individual detonation sites and  $16 \text{ g cm}^{-2}$  fuel loading, almost 40% of the BC is injected directly into the stratosphere. However, we show that direct injection of BC into the stratosphere makes little difference to the climate impact relative to upper tropospheric BC injections.

This study has illuminated the continued dependence of results on assumptions that remain uncertain and on models that remain underdeveloped for certain aspects of the problem. It is too early to predict with precision the global climate effects of the particular scenario selected for examination here. Moreover, these results are dependent on a single scenario and are not representative of the full range of possibilities for a regional nuclear exchange in South Asia or elsewhere. The large uncertainties lead to the conclusion that continued study is necessary. We must continue to test assumptions, become more effective at using existing modeling tools, and improve those tools, on a community-wide basis. Our future work on fires will likely include added complexity (such as use of WRF-Chem modules) and different burned-area shape or fire propagation. In EAM, we would like to develop more sophisticated smoke injection techniques, consider a wider range of BC particle size, include gases in the smoke plume (e.g., sulfur dioxide), and include more complete stratospheric chemistry. Finally, coupling EAM simulations to a dynamic ocean (fully coupled E3SM) are planned to resolve the surface temperature and precipitation anomalies and the dynamic ocean response.

### Data Availability Statement

ERA-Interim data are available free of charge (at <https://apps.ecmwf.int/datasets/>). The WRF model source code is archived at <https://doi.org/10.5065/D6MK6B4K> (Skamarock et al., 2019). The E3SM model was obtained from the Energy Exascale Earth System Model project, sponsored by the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research. The E3SM model source code is located in a publicly accessible repository: <https://doi.org/10.11578/E3SM/dc.20180418.36> (E3SM Project, 2018). The EAMv1 code used for the climate simulations can be obtained by retrieving the E3SM “maint-1.0” branch from the repository and configuring the model using the FC5AV1C compset. Note that this configuration does not contain the urban fire emissions modeled in WRF.



## Acknowledgments

This work was supported by LDRD project 18-ERD-049, "Modeling Nuclear Cloud Rise and Fallout in Complex Environments" and the LLNL Office of Defense Coordination. Data sets for this paper were created with the WRF and E3SM models. We would like to thank LLNL colleagues Craig Wuest for helpful discussions and Brad Roberts for improving this paper and helping place it in its historical context. Three-dimensional imagery for the fires (Figure 1) was produced by VAPOR ([www.vapor.ucar.edu](http://www.vapor.ucar.edu)), a product of the Computational Information Systems Laboratory at the National Center for Atmospheric Research. Prepared by LLNL under Contract DE-AC52-07NA27344.

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